

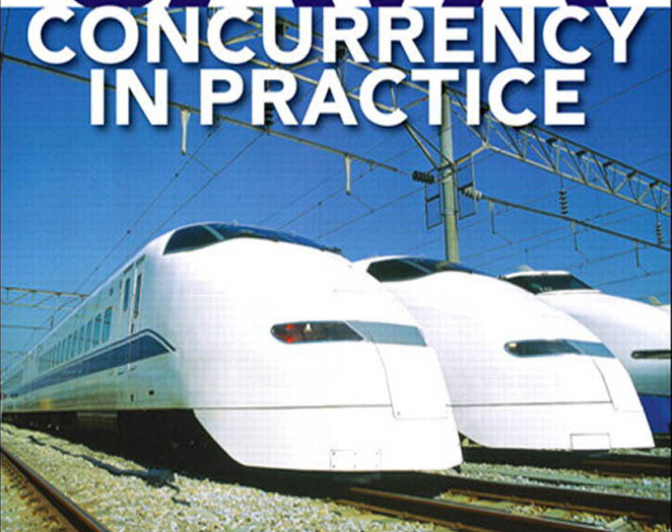
BRIAN GOETZ



WITH TIM PEIERLS, JOSHUA BLOCH,
JOSEPH BOWBEER, DAVID HOLMES,
AND DOUG LEA

JAVA

CONCURRENCY IN PRACTICE



FREE SAMPLE CHAPTER



SHARE WITH OTHERS

Advance praise for *Java Concurrency in Practice*

I was fortunate indeed to have worked with a fantastic team on the design and implementation of the concurrency features added to the Java platform in Java 5.0 and Java 6. Now this same team provides the best explanation yet of these new features, and of concurrency in general. Concurrency is no longer a subject for advanced users only. Every Java developer should read this book.

—Martin Buchholz
JDK Concurrency Czar, Sun Microsystems

For the past 30 years, computer performance has been driven by Moore's Law; from now on, it will be driven by Amdahl's Law. Writing code that effectively exploits multiple processors can be very challenging. *Java Concurrency in Practice* provides you with the concepts and techniques needed to write safe and scalable Java programs for today's—and tomorrow's—systems.

—Doron Rajwan
Research Scientist, Intel Corp

This is the book you need if you're writing—or designing, or debugging, or maintaining, or contemplating—multithreaded Java programs. If you've ever had to synchronize a method and you weren't sure why, you owe it to yourself and your users to read this book, cover to cover.

—Ted Neward
Author of Effective Enterprise Java

Brian addresses the fundamental issues and complexities of concurrency with uncommon clarity. This book is a must-read for anyone who uses threads and cares about performance.

—Kirk Pepperdine
CTO, JavaPerformanceTuning.com

This book covers a very deep and subtle topic in a very clear and concise way, making it the perfect Java Concurrency reference manual. Each page is filled with the problems (and solutions!) that programmers struggle with every day. Effectively exploiting concurrency is becoming more and more important now that Moore's Law is delivering more cores but not faster cores, and this book will show you how to do it.

—Dr. Cliff Click
Senior Software Engineer, Azul Systems

I have a strong interest in concurrency, and have probably written more thread deadlocks and made more synchronization mistakes than most programmers. Brian's book is the most readable on the topic of threading and concurrency in Java, and deals with this difficult subject with a wonderful hands-on approach. This is a book I am recommending to all my readers of The Java Specialists' Newsletter, because it is interesting, useful, and relevant to the problems facing Java developers today.

—Dr. Heinz Kabutz
The Java Specialists' Newsletter

I've focused a career on simplifying simple problems, but this book ambitiously and effectively works to simplify a complex but critical subject: concurrency. *Java Concurrency in Practice* is revolutionary in its approach, smooth and easy in style, and timely in its delivery—it's destined to be a very important book.

—Bruce Tate
Author of Beyond Java

Java Concurrency in Practice is an invaluable compilation of threading know-how for Java developers. I found reading this book intellectually exciting, in part because it is an excellent introduction to Java's concurrency API, but mostly because it captures in a thorough and accessible way expert knowledge on threading not easily found elsewhere.

—Bill Venners
Author of Inside the Java Virtual Machine

Java Concurrency in Practice

This page intentionally left blank

Java Concurrency in Practice

Brian Goetz
with
Tim Peierls
Joshua Bloch
Joseph Bowbeer
David Holmes
and Doug Lea

◆◆Addison-Wesley

Upper Saddle River, NJ • Boston • Indianapolis • San Francisco
New York • Toronto • Montreal • London • Munich • Paris • Madrid
Capetown • Sydney • Tokyo • Singapore • Mexico City

Many of the designations used by manufacturers and sellers to distinguish their products are claimed as trademarks. Where those designations appear in this book, and the publisher was aware of a trademark claim, the designations have been printed with initial capital letters or in all capitals.

The authors and publisher have taken care in the preparation of this book, but make no expressed or implied warranty of any kind and assume no responsibility for errors or omissions. No liability is assumed for incidental or consequential damages in connection with or arising out of the use of the information or programs contained herein.

The publisher offers excellent discounts on this book when ordered in quantity for bulk purchases or special sales, which may include electronic versions and/or custom covers and content particular to your business, training goals, marketing focus, and branding interests. For more information, please contact:

U.S. Corporate and Government Sales
(800) 382-3419
corpsales@pearsontechgroup.com

For sales outside the United States, please contact:

International Sales
international@pearsoned.com

Visit us on the Web: www.awprofessional.com



This Book Is Safari Enabled

The Safari® Enabled icon on the cover of your favorite technology book means the book is available through Safari Bookshelf. When you buy this book, you get free access to the online edition for 45 days.

Safari Bookshelf is an electronic reference library that lets you easily search thousands of technical books, find code samples, download chapters, and access technical information whenever and wherever you need it.

To gain 45-day Safari Enabled access to this book:

- Go to <http://www.awprofessional.com/safarienabled>
- Complete the brief registration form
- Enter the coupon code

If you have difficulty registering on Safari Bookshelf or accessing the online edition, please e-mail customer-service@safaribooksonline.com.

Library of Congress Cataloging-in-Publication Data

Goetz, Brian.

Java Concurrency in Practice / Brian Goetz, with Tim Peierls. . . [et al.]
p. cm.

Includes bibliographical references and index.

ISBN 0-321-34960-1 (pbk. : alk. paper)

1. Java (Computer program language) 2. Parallel programming (Computer science) 3. Threads (Computer programs) I. Title.

QA76.73.J38G588 2006
005.13'3--dc22

2006012205

Copyright © 2006 Pearson Education, Inc.

ISBN 0-321-34960-1

Text printed in the United States on recycled paper at Courier Stoughton in Stoughton, Massachusetts.

13th Printing

To Jessica

This page intentionally left blank

Contents

Listings	xii
Preface	xvii
1 Introduction	1
1.1 A (very) brief history of concurrency	1
1.2 Benefits of threads	3
1.3 Risks of threads	5
1.4 Threads are everywhere	9
I Fundamentals	13
2 Thread Safety	15
2.1 What is thread safety?	17
2.2 Atomicity	19
2.3 Locking	23
2.4 Guarding state with locks	27
2.5 Liveness and performance	29
3 Sharing Objects	33
3.1 Visibility	33
3.2 Publication and escape	39
3.3 Thread confinement	42
3.4 Immutability	46
3.5 Safe publication	49
4 Composing Objects	55
4.1 Designing a thread-safe class	55
4.2 Instance confinement	58
4.3 Delegating thread safety	62
4.4 Adding functionality to existing thread-safe classes	71
4.5 Documenting synchronization policies	74

5	Building Blocks	79
5.1	Synchronized collections	79
5.2	Concurrent collections	84
5.3	Blocking queues and the producer-consumer pattern	87
5.4	Blocking and interruptible methods	92
5.5	Synchronizers	94
5.6	Building an efficient, scalable result cache	101
II	Structuring Concurrent Applications	111
6	Task Execution	113
6.1	Executing tasks in threads	113
6.2	The Executor framework	117
6.3	Finding exploitable parallelism	123
7	Cancellation and Shutdown	135
7.1	Task cancellation	135
7.2	Stopping a thread-based service	150
7.3	Handling abnormal thread termination	161
7.4	JVM shutdown	164
8	Applying Thread Pools	167
8.1	Implicit couplings between tasks and execution policies	167
8.2	Sizing thread pools	170
8.3	Configuring ThreadPoolExecutor	171
8.4	Extending ThreadPoolExecutor	179
8.5	Parallelizing recursive algorithms	181
9	GUI Applications	189
9.1	Why are GUIs single-threaded?	189
9.2	Short-running GUI tasks	192
9.3	Long-running GUI tasks	195
9.4	Shared data models	198
9.5	Other forms of single-threaded subsystems	202
III	Liveness, Performance, and Testing	203
10	Avoiding Liveness Hazards	205
10.1	Deadlock	205
10.2	Avoiding and diagnosing deadlocks	215
10.3	Other liveness hazards	218
11	Performance and Scalability	221
11.1	Thinking about performance	221
11.2	Amdahl's law	225
11.3	Costs introduced by threads	229
11.4	Reducing lock contention	232

11.5	Example: Comparing Map performance	242
11.6	Reducing context switch overhead	243
12	Testing Concurrent Programs	247
12.1	Testing for correctness	248
12.2	Testing for performance	260
12.3	Avoiding performance testing pitfalls	266
12.4	Complementary testing approaches	270
IV	Advanced Topics	275
13	Explicit Locks	277
13.1	Lock and ReentrantLock	277
13.2	Performance considerations	282
13.3	Fairness	283
13.4	Choosing between synchronized and ReentrantLock	285
13.5	Read-write locks	286
14	Building Custom Synchronizers	291
14.1	Managing state dependence	291
14.2	Using condition queues	298
14.3	Explicit condition objects	306
14.4	Anatomy of a synchronizer	308
14.5	AbstractQueuedSynchronizer	311
14.6	AQS in java.util.concurrent synchronizer classes	314
15	Atomic Variables and Nonblocking Synchronization	319
15.1	Disadvantages of locking	319
15.2	Hardware support for concurrency	321
15.3	Atomic variable classes	324
15.4	Nonblocking algorithms	329
16	The Java Memory Model	337
16.1	What is a memory model, and why would I want one?	337
16.2	Publication	344
16.3	Initialization safety	349
A	Annotations for Concurrency	353
A.1	Class annotations	353
A.2	Field and method annotations	353
	Bibliography	355
	Index	359

Listings

1	Bad way to sort a list. <i>Don't do this.</i>	xix
2	Less than optimal way to sort a list.	xx
1.1	Non-thread-safe sequence generator.	6
1.2	Thread-safe sequence generator.	7
2.1	A stateless servlet.	18
2.2	Servlet that counts requests without the necessary synchronization. <i>Don't do this.</i>	19
2.3	Race condition in lazy initialization. <i>Don't do this.</i>	21
2.4	Servlet that counts requests using <code>AtomicLong</code>	23
2.5	Servlet that attempts to cache its last result without adequate atomicity. <i>Don't do this.</i>	24
2.6	Servlet that caches last result, but with unacceptably poor concurrency. <i>Don't do this.</i>	26
2.7	Code that would deadlock if intrinsic locks were not reentrant.	27
2.8	Servlet that caches its last request and result.	31
3.1	Sharing variables without synchronization. <i>Don't do this.</i>	34
3.2	Non-thread-safe mutable integer holder.	36
3.3	Thread-safe mutable integer holder.	36
3.4	Counting sheep.	39
3.5	Publishing an object.	40
3.6	Allowing internal mutable state to escape. <i>Don't do this.</i>	40
3.7	Implicitly allowing the <code>this</code> reference to escape. <i>Don't do this.</i>	41
3.8	Using a factory method to prevent the <code>this</code> reference from escaping during construction.	42
3.9	Thread confinement of local primitive and reference variables.	44
3.10	Using <code>ThreadLocal</code> to ensure thread confinement.	45
3.11	Immutable class built out of mutable underlying objects.	47
3.12	Immutable holder for caching a number and its factors.	49
3.13	Caching the last result using a volatile reference to an immutable holder object.	50
3.14	Publishing an object without adequate synchronization. <i>Don't do this.</i>	50
3.15	Class at risk of failure if not properly published.	51
4.1	Simple thread-safe counter using the Java monitor pattern.	56
4.2	Using confinement to ensure thread safety.	59
4.3	Guarding state with a private lock.	61

4.4	Monitor-based vehicle tracker implementation.	63
4.5	Mutable point class similar to <code>java.awt.Point</code>	64
4.6	Immutable <code>Point</code> class used by <code>DelegatingVehicleTracker</code>	64
4.7	Delegating thread safety to a <code>ConcurrentHashMap</code>	65
4.8	Returning a static copy of the location set instead of a “live” one.	66
4.9	Delegating thread safety to multiple underlying state variables.	66
4.10	Number range class that does not sufficiently protect its invariants. <i>Don't do this.</i>	67
4.11	Thread-safe mutable point class.	69
4.12	Vehicle tracker that safely publishes underlying state.	70
4.13	Extending <code>Vector</code> to have a put-if-absent method.	72
4.14	Non-thread-safe attempt to implement put-if-absent. <i>Don't do this.</i>	72
4.15	Implementing put-if-absent with client-side locking.	73
4.16	Implementing put-if-absent using composition.	74
5.1	Compound actions on a <code>Vector</code> that may produce confusing results.	80
5.2	Compound actions on <code>Vector</code> using client-side locking.	81
5.3	Iteration that may throw <code>ArrayIndexOutOfBoundsException</code>	81
5.4	Iteration with client-side locking.	82
5.5	Iterating a <code>List</code> with an <code>Iterator</code>	82
5.6	Iteration hidden within string concatenation. <i>Don't do this.</i>	84
5.7	<code>ConcurrentMap</code> interface.	87
5.8	Producer and consumer tasks in a desktop search application.	91
5.9	Starting the desktop search.	92
5.10	Restoring the interrupted status so as not to swallow the interrupt.	94
5.11	Using <code>CountDownLatch</code> for starting and stopping threads in timing tests.	96
5.12	Using <code>FutureTask</code> to preload data that is needed later.	97
5.13	Coercing an unchecked <code>Throwable</code> to a <code>RuntimeException</code>	98
5.14	Using <code>Semaphore</code> to bound a collection.	100
5.15	Coordinating computation in a cellular automaton with <code>CyclicBarrier</code>	102
5.16	Initial cache attempt using <code>HashMap</code> and synchronization.	103
5.17	Replacing <code>HashMap</code> with <code>ConcurrentHashMap</code>	105
5.18	Memoizing wrapper using <code>FutureTask</code>	106
5.19	Final implementation of <code>Memoizer</code>	108
5.20	Factorizing servlet that caches results using <code>Memoizer</code>	109
6.1	Sequential web server.	114
6.2	Web server that starts a new thread for each request.	115
6.3	<code>Executor</code> interface.	117
6.4	Web server using a thread pool.	118
6.5	<code>Executor</code> that starts a new thread for each task.	118
6.6	<code>Executor</code> that executes tasks synchronously in the calling thread.	119
6.7	Lifecycle methods in <code>ExecutorService</code>	121
6.8	Web server with shutdown support.	122
6.9	Class illustrating confusing <code>Timer</code> behavior.	124
6.10	Rendering page elements sequentially.	125
6.11	<code>Callable</code> and <code>Future</code> interfaces.	126

6.12	Default implementation of <code>newTaskFor</code> in <code>ThreadPoolExecutor</code> . . .	126
6.13	Waiting for image download with <code>Future</code>	128
6.14	<code>QueueingFuture</code> class used by <code>ExecutorCompletionService</code>	129
6.15	Using <code>CompletionService</code> to render page elements as they become available.	130
6.16	Fetching an advertisement with a time budget.	132
6.17	Requesting travel quotes under a time budget.	134
7.1	Using a <code>volatile</code> field to hold cancellation state.	137
7.2	Generating a second's worth of prime numbers.	137
7.3	Unreliable cancellation that can leave producers stuck in a blocking operation. <i>Don't do this</i>	139
7.4	Interruption methods in <code>Thread</code>	139
7.5	Using interruption for cancellation.	141
7.6	Propagating <code>InterruptedException</code> to callers.	143
7.7	Noncancelable task that restores interruption before exit.	144
7.8	Scheduling an interrupt on a borrowed thread. <i>Don't do this</i>	145
7.9	Interrupting a task in a dedicated thread.	146
7.10	Cancelling a task using <code>Future</code>	147
7.11	Encapsulating nonstandard cancellation in a <code>Thread</code> by overriding <code>interrupt</code>	149
7.12	Encapsulating nonstandard cancellation in a task with <code>newTaskFor</code> . .	151
7.13	Producer-consumer logging service with no shutdown support. . .	152
7.14	Unreliable way to add shutdown support to the logging service. . .	153
7.15	Adding reliable cancellation to <code>LogWriter</code>	154
7.16	Logging service that uses an <code>ExecutorService</code>	155
7.17	Shutdown with poison pill.	156
7.18	Producer thread for <code>IndexingService</code>	157
7.19	Consumer thread for <code>IndexingService</code>	157
7.20	Using a private <code>Executor</code> whose lifetime is bounded by a method call.	158
7.21	<code>ExecutorService</code> that keeps track of cancelled tasks after shutdown. .	159
7.22	Using <code>TrackingExecutorService</code> to save unfinished tasks for later execution.	160
7.23	Typical thread-pool worker thread structure.	162
7.24	<code>UncaughtExceptionHandler</code> interface.	163
7.25	<code>UncaughtExceptionHandler</code> that logs the exception.	163
7.26	Registering a shutdown hook to stop the logging service.	165
8.1	Task that deadlocks in a single-threaded <code>Executor</code> . <i>Don't do this</i> . .	169
8.2	General constructor for <code>ThreadPoolExecutor</code>	172
8.3	Creating a fixed-sized thread pool with a bounded queue and the caller-runs saturation policy.	175
8.4	Using a <code>Semaphore</code> to throttle task submission.	176
8.5	<code>ThreadFactory</code> interface.	176
8.6	Custom thread factory.	177
8.7	Custom thread base class.	178
8.8	Modifying an <code>Executor</code> created with the standard factories. . . .	179
8.9	Thread pool extended with logging and timing.	180

8.10	Transforming sequential execution into parallel execution.	181
8.11	Transforming sequential tail-recursion into parallelized recursion. . .	182
8.12	Waiting for results to be calculated in parallel.	182
8.13	Abstraction for puzzles like the “sliding blocks puzzle”.	183
8.14	Link node for the puzzle solver framework.	184
8.15	Sequential puzzle solver.	185
8.16	Concurrent version of puzzle solver.	186
8.17	Result-bearing latch used by ConcurrentPuzzleSolver.	187
8.18	Solver that recognizes when no solution exists.	188
9.1	Implementing SwingUtilities using an Executor.	193
9.2	Executor built atop SwingUtilities.	194
9.3	Simple event listener.	194
9.4	Binding a long-running task to a visual component.	196
9.5	Long-running task with user feedback.	196
9.6	Cancelling a long-running task.	197
9.7	Background task class supporting cancellation, completion notification, and progress notification.	199
9.8	Initiating a long-running, cancellable task with BackgroundTask. . .	200
10.1	Simple lock-ordering deadlock. <i>Don't do this.</i>	207
10.2	Dynamic lock-ordering deadlock. <i>Don't do this.</i>	208
10.3	Inducing a lock ordering to avoid deadlock.	209
10.4	Driver loop that induces deadlock under typical conditions.	210
10.5	Lock-ordering deadlock between cooperating objects. <i>Don't do this.</i> .	212
10.6	Using open calls to avoiding deadlock between cooperating objects. .	214
10.7	Portion of thread dump after deadlock.	217
11.1	Serialized access to a task queue.	227
11.2	Synchronization that has no effect. <i>Don't do this.</i>	230
11.3	Candidate for lock elision.	231
11.4	Holding a lock longer than necessary.	233
11.5	Reducing lock duration.	234
11.6	Candidate for lock splitting.	236
11.7	ServerStatus refactored to use split locks.	236
11.8	Hash-based map using lock striping.	238
12.1	Bounded buffer using Semaphore.	249
12.2	Basic unit tests for BoundedBuffer.	250
12.3	Testing blocking and responsiveness to interruption.	252
12.4	Medium-quality random number generator suitable for testing. . .	253
12.5	Producer-consumer test program for BoundedBuffer.	255
12.6	Producer and consumer classes used in PutTakeTest.	256
12.7	Testing for resource leaks.	258
12.8	Thread factory for testing ThreadPoolExecutor.	258
12.9	Test method to verify thread pool expansion.	259
12.10	Using Thread.yield to generate more interleavings.	260
12.11	Barrier-based timer.	261
12.12	Testing with a barrier-based timer.	262
12.13	Driver program for TimedPutTakeTest.	262
13.1	Lock interface.	277

13.2	Guarding object state using <code>ReentrantLock</code>	278
13.3	Avoiding lock-ordering deadlock using <code>tryLock</code>	280
13.4	Locking with a time budget.	281
13.5	Interruptible lock acquisition.	281
13.6	<code>ReadWriteLock</code> interface.	286
13.7	Wrapping a <code>Map</code> with a read-write lock.	288
14.1	Structure of blocking state-dependent actions.	292
14.2	Base class for bounded buffer implementations.	293
14.3	Bounded buffer that balks when preconditions are not met.	294
14.4	Client logic for calling <code>GrumpyBoundedBuffer</code>	294
14.5	Bounded buffer using crude blocking.	296
14.6	Bounded buffer using condition queues.	298
14.7	Canonical form for state-dependent methods.	301
14.8	Using conditional notification in <code>BoundedBuffer.put</code>	304
14.9	Recloseable gate using <code>wait</code> and <code>notifyAll</code>	305
14.10	Condition interface.	307
14.11	Bounded buffer using explicit condition variables.	309
14.12	Counting semaphore implemented using <code>Lock</code>	310
14.13	Canonical forms for acquisition and release in <code>AQS</code>	312
14.14	Binary latch using <code>AbstractQueuedSynchronizer</code>	313
14.15	<code>tryAcquire</code> implementation from nonfair <code>ReentrantLock</code>	315
14.16	<code>tryAcquireShared</code> and <code>tryReleaseShared</code> from <code>Semaphore</code>	316
15.1	Simulated CAS operation.	322
15.2	Nonblocking counter using CAS.	323
15.3	Preserving multivariable invariants using CAS.	326
15.4	Random number generator using <code>ReentrantLock</code>	327
15.5	Random number generator using <code>AtomicInteger</code>	327
15.6	Nonblocking stack using Treiber's algorithm (Treiber, 1986).	331
15.7	Insertion in the Michael-Scott nonblocking queue algorithm (Michael and Scott, 1996).	334
15.8	Using atomic field updaters in <code>ConcurrentLinkedQueue</code>	335
16.1	Insufficiently synchronized program that can have surprising re- sults. <i>Don't do this</i>	340
16.2	Inner class of <code>FutureTask</code> illustrating synchronization piggybacking.	343
16.3	Unsafe lazy initialization. <i>Don't do this</i>	345
16.4	Thread-safe lazy initialization.	347
16.5	Eager initialization.	347
16.6	Lazy initialization holder class idiom.	348
16.7	Double-checked-locking antipattern. <i>Don't do this</i>	349
16.8	Initialization safety for immutable objects.	350

Preface

At this writing, multicore processors are just now becoming inexpensive enough for midrange desktop systems. Not coincidentally, many development teams are noticing more and more threading-related bug reports in their projects. In a recent post on the NetBeans developer site, one of the core maintainers observed that a single class had been patched over 14 times to fix threading-related problems. Dion Almaer, former editor of TheServerSide, recently blogged (after a painful debugging session that ultimately revealed a threading bug) that most Java programs are so rife with concurrency bugs that they work only “by accident”.

Indeed, developing, testing and debugging multithreaded programs can be extremely difficult because concurrency bugs do not manifest themselves predictably. And when they do surface, it is often at the worst possible time—in production, under heavy load.

One of the challenges of developing concurrent programs in Java is the mismatch between the concurrency features offered by the platform and how developers need to think about concurrency in their programs. The language provides low-level *mechanisms* such as synchronization and condition waits, but these mechanisms must be used consistently to implement application-level protocols or *policies*. Without such policies, it is all too easy to create programs that compile and appear to work but are nevertheless broken. Many otherwise excellent books on concurrency fall short of their goal by focusing excessively on low-level mechanisms and APIs rather than design-level policies and patterns.

Java 5.0 is a huge step forward for the development of concurrent applications in Java, providing new higher-level components and additional low-level mechanisms that make it easier for novices and experts alike to build concurrent applications. The authors are the primary members of the JCP Expert Group that created these facilities; in addition to describing their behavior and features, we present the underlying design patterns and anticipated usage scenarios that motivated their inclusion in the platform libraries.

Our goal is to give readers a set of design rules and mental models that make it easier—and more fun—to build correct, performant concurrent classes and applications in Java.

We hope you enjoy *Java Concurrency in Practice*.

Brian Goetz
Williston, VT
March 2006

How to use this book

To address the abstraction mismatch between Java’s low-level mechanisms and the necessary design-level policies, we present a *simplified* set of rules for writing concurrent programs. Experts may look at these rules and say “Hmm, that’s not entirely true: class *C* is thread-safe even though it violates rule *R*.” While it is possible to write correct programs that break our rules, doing so requires a deep understanding of the low-level details of the Java Memory Model, and we want developers to be able to write correct concurrent programs *without* having to master these details. Consistently following our simplified rules will produce correct and maintainable concurrent programs.

We assume the reader already has some familiarity with the basic mechanisms for concurrency in Java. *Java Concurrency in Practice* is not an introduction to concurrency—for that, see the threading chapter of any decent introductory volume, such as *The Java Programming Language* (Arnold et al., 2005). Nor is it an encyclopedic reference for All Things Concurrency—for that, see *Concurrent Programming in Java* (Lea, 2000). Rather, it offers practical design rules to assist developers in the difficult process of creating safe and performant concurrent classes. Where appropriate, we cross-reference relevant sections of *The Java Programming Language*, *Concurrent Programming in Java*, *The Java Language Specification* (Gosling et al., 2005), and *Effective Java* (Bloch, 2001) using the conventions [JPL n.m], [CPJ n.m], [JLS n.m], and [EJ Item n].

After the introduction (Chapter 1), the book is divided into four parts:

Fundamentals. Part I (Chapters 2-5) focuses on the basic concepts of concurrency and thread safety, and how to compose thread-safe classes out of the concurrent building blocks provided by the class library. A “cheat sheet” summarizing the most important of the rules presented in Part I appears on page 110.

Chapters 2 (Thread Safety) and 3 (Sharing Objects) form the foundation for the book. Nearly all of the rules on avoiding concurrency hazards, constructing thread-safe classes, and verifying thread safety are here. Readers who prefer “practice” to “theory” may be tempted to skip ahead to Part II, but make sure to come back and read Chapters 2 and 3 before writing any concurrent code!

Chapter 4 (Composing Objects) covers techniques for composing thread-safe classes into larger thread-safe classes. Chapter 5 (Building Blocks) covers the concurrent building blocks—thread-safe collections and synchronizers—provided by the platform libraries.

Structuring Concurrent Applications. Part II (Chapters 6-9) describes how to exploit threads to improve the throughput or responsiveness of concurrent applications. Chapter 6 (Task Execution) covers identifying parallelizable tasks and executing them within the task-execution framework. Chapter 7 (Cancellation and Shutdown) deals with techniques for convincing tasks and threads to terminate before they would normally do so; how programs deal with cancellation and shutdown is often one of the factors that separates truly robust concurrent applications from those that merely work. Chapter 8 (Applying Thread Pools) addresses some of the more advanced features of the task-execution framework.

Chapter 9 (GUI Applications) focuses on techniques for improving responsiveness in single-threaded subsystems.

Liveness, Performance, and Testing. Part III (Chapters 10-12) concerns itself with ensuring that concurrent programs actually do what you want them to do and do so with acceptable performance. Chapter 10 (Avoiding Liveness Hazards) describes how to avoid liveness failures that can prevent programs from making forward progress. Chapter 11 (Performance and Scalability) covers techniques for improving the performance and scalability of concurrent code. Chapter 12 (Testing Concurrent Programs) covers techniques for testing concurrent code for both correctness and performance.

Advanced Topics. Part IV (Chapters 13-16) covers topics that are likely to be of interest only to experienced developers: explicit locks, atomic variables, nonblocking algorithms, and developing custom synchronizers.

Code examples

While many of the general concepts in this book are applicable to versions of Java prior to Java 5.0 and even to non-Java environments, most of the code examples (and all the statements about the Java Memory Model) assume Java 5.0 or later. Some of the code examples may use library features added in Java 6.

The code examples have been compressed to reduce their size and to highlight the relevant portions. The full versions of the code examples, as well as supplementary examples and errata, are available from the book's website, <http://www.javaconcurrencyinpractice.com>.

The code examples are of three sorts: “good” examples, “not so good” examples, and “bad” examples. Good examples illustrate techniques that should be emulated. Bad examples illustrate techniques that should definitely *not* be emulated, and are identified with a “Mr. Yuk” icon¹ to make it clear that this is “toxic” code (see Listing 1). Not-so-good examples illustrate techniques that are not *necessarily* wrong but are fragile, risky, or perform poorly, and are decorated with a “Mr. Could Be Happier” icon as in Listing 2.

```
public <T extends Comparable<? super T>> void sort(List<T> list) {
    // Never returns the wrong answer!
    System.exit(0);
}
```



LISTING 1. Bad way to sort a list. *Don't do this.*

Some readers may question the role of the “bad” examples in this book; after all, a book should show how to do things right, not wrong. The bad examples have two purposes. They illustrate common pitfalls, but more importantly they demonstrate how to analyze a program for thread safety—and the best way to do that is to see the ways in which thread safety is compromised.

1. Mr. Yuk is a registered trademark of the Children's Hospital of Pittsburgh and appears by permission.

```
public <T extends Comparable<? super T>> void sort(List<T> list) {
    for (int i=0; i<1000000; i++)
        doNothing();
    Collections.sort(list);
}
```



LISTING 2. Less than optimal way to sort a list.

Acknowledgments

This book grew out of the development process for the `java.util.concurrent` package that was created by the Java Community Process JSR 166 for inclusion in Java 5.0. Many others contributed to JSR 166; in particular we thank Martin Buchholz for doing all the work related to getting the code into the JDK, and all the readers of the concurrency-interest mailing list who offered their suggestions and feedback on the draft APIs.

This book has been tremendously improved by the suggestions and assistance of a small army of reviewers, advisors, cheerleaders, and armchair critics. We would like to thank Dion Almaer, Tracy Bialik, Cindy Bloch, Martin Buchholz, Paul Christmann, Cliff Click, Stuart Halloway, David Hovemeyer, Jason Hunter, Michael Hunter, Jeremy Hylton, Heinz Kabutz, Robert Kuhar, Ramnivas Laddad, Jared Levy, Nicole Lewis, Victor Luchangco, Jeremy Manson, Paul Martin, Berna Massingill, Michael Maurer, Ted Neward, Kirk Pepperdine, Bill Pugh, Sam Pullara, Russ Rufer, Bill Scherer, Jeffrey Siegal, Bruce Tate, Gil Tene, Paul Tyma, and members of the Silicon Valley Patterns Group who, through many interesting technical conversations, offered guidance and made suggestions that helped make this book better.

We are especially grateful to Cliff Biffle, Barry Hayes, Dawid Kurzyniec, Angelika Langer, Doron Rajwan, and Bill Venners, who reviewed the entire manuscript in excruciating detail, found bugs in the code examples, and suggested numerous improvements.

We thank Katrina Avery for a great copy-editing job and Rosemary Simpson for producing the index under unreasonable time pressure. We thank Ami Dewar for doing the illustrations.

Thanks to the whole team at Addison-Wesley who helped make this book a reality. Ann Sellers got the project launched and Greg Doench shepherded it to a smooth completion; Elizabeth Ryan guided it through the production process.

We would also like to thank the thousands of software engineers who contributed indirectly by creating the software used to create this book, including $\text{\TeX}$, $\text{\LaTeX}$, Adobe Acrobat, `pic`, `grap`, Adobe Illustrator, Perl, Apache Ant, IntelliJ IDEA, GNU emacs, Subversion, TortoiseSVN, and of course, the Java platform and class libraries.

This page intentionally left blank

CHAPTER 6

Task Execution

Most concurrent applications are organized around the execution of *tasks*: abstract, discrete units of work. Dividing the work of an application into tasks simplifies program organization, facilitates error recovery by providing natural transaction boundaries, and promotes concurrency by providing a natural structure for parallelizing work.

6.1 Executing tasks in threads

The first step in organizing a program around task execution is identifying sensible *task boundaries*. Ideally, tasks are *independent* activities: work that doesn't depend on the state, result, or side effects of other tasks. Independence facilitates concurrency, as independent tasks can be executed in parallel if there are adequate processing resources. For greater flexibility in scheduling and load balancing tasks, each task should also represent a small fraction of your application's processing capacity.

Server applications should exhibit both *good throughput* and *good responsiveness* under normal load. Application providers want applications to support as many users as possible, so as to reduce provisioning costs per user; users want to get their response quickly. Further, applications should exhibit *graceful degradation* as they become overloaded, rather than simply falling over under heavy load. Choosing good task boundaries, coupled with a sensible *task execution policy* (see Section 6.2.2), can help achieve these goals.

Most server applications offer a natural choice of task boundary: individual client requests. Web servers, mail servers, file servers, EJB containers, and database servers all accept requests via network connections from remote clients. Using individual requests as task boundaries usually offers both independence and appropriate task sizing. For example, the result of submitting a message to a mail server is not affected by the other messages being processed at the same time, and handling a single message usually requires a very small percentage of the server's total capacity.

6.1.1 Executing tasks sequentially

There are a number of possible policies for scheduling tasks within an application, some of which exploit the potential for concurrency better than others. The simplest is to execute tasks sequentially in a single thread. `SingleThreadWebServer` in Listing 6.1 processes its tasks—HTTP requests arriving on port 80—sequentially. The details of the request processing aren't important; we're interested in characterizing the concurrency of various scheduling policies.

```
class SingleThreadWebServer {  
    public static void main(String[] args) throws IOException {  
        ServerSocket socket = new ServerSocket(80);  
        while (true) {  
            Socket connection = socket.accept();  
            handleRequest(connection);  
        }  
    }  
}
```



LISTING 6.1. Sequential web server.

`SingleThreadedWebServer` is simple and theoretically correct, but would perform poorly in production because it can handle only one request at a time. The main thread alternates between accepting connections and processing the associated request. While the server is handling a request, new connections must wait until it finishes the current request and calls `accept` again. This might work if request processing were so fast that `handleRequest` effectively returned immediately, but this doesn't describe any web server in the real world.

Processing a web request involves a mix of computation and I/O. The server must perform socket I/O to read the request and write the response, which can block due to network congestion or connectivity problems. It may also perform file I/O or make database requests, which can also block. In a single-threaded server, blocking not only delays completing the current request, but prevents pending requests from being processed at all. If one request blocks for an unusually long time, users might think the server is unavailable because it appears unresponsive. At the same time, resource utilization is poor, since the CPU sits idle while the single thread waits for its I/O to complete.

In server applications, sequential processing rarely provides either good throughput or good responsiveness. There are exceptions—such as when tasks are few and long-lived, or when the server serves a single client that makes only a single request at a time—but most server applications do not work this way.¹

1. In some situations, sequential processing may offer a simplicity or safety advantage; most GUI frameworks process tasks sequentially using a single thread. We return to the sequential model in Chapter 9.

6.1.2 Explicitly creating threads for tasks

A more responsive approach is to create a new thread for servicing each request, as shown in `ThreadPerTaskWebServer` in Listing 6.2.

```
class ThreadPerTaskWebServer {  
    public static void main(String[] args) throws IOException {  
        ServerSocket socket = new ServerSocket(80);  
        while (true) {  
            final Socket connection = socket.accept();  
            Runnable task = new Runnable() {  
                public void run() {  
                    handleRequest(connection);  
                }  
            };  
            new Thread(task).start();  
        }  
    }  
}
```



LISTING 6.2. Web server that starts a new thread for each request.

`ThreadPerTaskWebServer` is similar in structure to the single-threaded version—the main thread still alternates between accepting an incoming connection and dispatching the request. The difference is that for each connection, the main loop creates a new thread to process the request instead of processing it within the main thread. This has three main consequences:

- Task processing is offloaded from the main thread, enabling the main loop to resume waiting for the next incoming connection more quickly. This enables new connections to be accepted before previous requests complete, improving responsiveness.
- Tasks can be processed in parallel, enabling multiple requests to be serviced simultaneously. This may improve throughput if there are multiple processors, or if tasks need to block for any reason such as I/O completion, lock acquisition, or resource availability.
- Task-handling code must be thread-safe, because it may be invoked concurrently for multiple tasks.

Under light to moderate load, the thread-per-task approach is an improvement over sequential execution. As long as the request arrival rate does not exceed the server's capacity to handle requests, this approach offers better responsiveness and throughput.

6.1.3 Disadvantages of unbounded thread creation

For production use, however, the thread-per-task approach has some practical drawbacks, especially when a large number of threads may be created:

Thread lifecycle overhead. Thread creation and teardown are not free. The actual overhead varies across platforms, but thread creation takes time, introducing latency into request processing, and requires some processing activity by the JVM and OS. If requests are frequent and lightweight, as in most server applications, creating a new thread for each request can consume significant computing resources.

Resource consumption. Active threads consume system resources, especially memory. When there are more runnable threads than available processors, threads sit idle. Having many idle threads can tie up a lot of memory, putting pressure on the garbage collector, and having many threads competing for the CPUs can impose other performance costs as well. If you have enough threads to keep all the CPUs busy, creating more threads won't help and may even hurt.

Stability. There is a limit on how many threads can be created. The limit varies by platform and is affected by factors including JVM invocation parameters, the requested stack size in the Thread constructor, and limits on threads placed by the underlying operating system.² When you hit this limit, the most likely result is an `OutOfMemoryError`. Trying to recover from such an error is very risky; it is far easier to structure your program to avoid hitting this limit.

Up to a certain point, more threads can improve throughput, but beyond that point creating more threads just slows down your application, and creating one thread too many can cause your entire application to crash horribly. The way to stay out of danger is to place some bound on how many threads your application creates, and to test your application thoroughly to ensure that, even when this bound is reached, it does not run out of resources.

The problem with the thread-per-task approach is that nothing places any limit on the number of threads created except the rate at which remote users can throw HTTP requests at it. Like other concurrency hazards, unbounded thread creation may *appear* to work just fine during prototyping and development, with problems surfacing only when the application is deployed and under heavy load. So a malicious user, or enough ordinary users, can make your web server crash if the traffic load ever reaches a certain threshold. For a server application that is supposed to provide high availability and graceful degradation under load, this is a serious failing.

2. On 32-bit machines, a major limiting factor is address space for thread stacks. Each thread maintains two execution stacks, one for Java code and one for native code. Typical JVM defaults yield a combined stack size of around half a megabyte. (You can change this with the `-Xss` JVM flag or through the Thread constructor.) If you divide the per-thread stack size into 2^{32} , you get a limit of a few thousands or tens of thousands of threads. Other factors, such as OS limitations, may impose stricter limits.

6.2 The Executor framework

Tasks are logical units of work, and threads are a mechanism by which tasks can run asynchronously. We've examined two policies for executing tasks using threads—execute tasks sequentially in a single thread, and execute each task in its own thread. Both have serious limitations: the sequential approach suffers from poor responsiveness and throughput, and the thread-per-task approach suffers from poor resource management.

In Chapter 5, we saw how to use *bounded queues* to prevent an overloaded application from running out of memory. *Thread pools* offer the same benefit for thread management, and `java.util.concurrent` provides a flexible thread pool implementation as part of the Executor framework. The primary abstraction for task execution in the Java class libraries is *not* `Thread`, but `Executor`, shown in Listing 6.3.

```
public interface Executor {  
    void execute(Runnable command);  
}
```

LISTING 6.3. Executor interface.

`Executor` may be a simple interface, but it forms the basis for a flexible and powerful framework for asynchronous task execution that supports a wide variety of task execution policies. It provides a standard means of decoupling *task submission* from *task execution*, describing tasks with `Runnable`. The `Executor` implementations also provide lifecycle support and hooks for adding statistics gathering, application management, and monitoring.

`Executor` is based on the producer-consumer pattern, where activities that submit tasks are the producers (producing units of work to be done) and the threads that execute tasks are the consumers (consuming those units of work). *Using an Executor is usually the easiest path to implementing a producer-consumer design in your application.*

6.2.1 Example: web server using Executor

Building a web server with an `Executor` is easy. `TaskExecutionWebServer` in Listing 6.4 replaces the hard-coded thread creation with an `Executor`. In this case, we use one of the standard `Executor` implementations, a fixed-size thread pool with 100 threads.

In `TaskExecutionWebServer`, submission of the request-handling task is decoupled from its execution using an `Executor`, and its behavior can be changed merely by substituting a different `Executor` implementation. Changing `Executor` implementations or configuration is far less invasive than changing the way tasks are submitted; `Executor` configuration is generally a one-time event and can easily be exposed for deployment-time configuration, whereas task submission code tends to be strewn throughout the program and harder to expose.

```

class TaskExecutionWebServer {
    private static final int NTHREADS = 100;
    private static final Executor exec
        = Executors.newFixedThreadPool(NTHREADS);

    public static void main(String[] args) throws IOException {
        ServerSocket socket = new ServerSocket(80);
        while (true) {
            final Socket connection = socket.accept();
            Runnable task = new Runnable() {
                public void run() {
                    handleRequest(connection);
                }
            };
            exec.execute(task);
        }
    }
}

```

LISTING 6.4. Web server using a thread pool.

We can easily modify `TaskExecutionWebServer` to behave like `ThreadPerTaskWebServer` by substituting an `Executor` that creates a new thread for each request. Writing such an `Executor` is trivial, as shown in `ThreadPerTaskExecutor` in Listing 6.5.

```

public class ThreadPerTaskExecutor implements Executor {
    public void execute(Runnable r) {
        new Thread(r).start();
    }
}

```

LISTING 6.5. Executor that starts a new thread for each task.

Similarly, it is also easy to write an `Executor` that would make `TaskExecutionWebServer` behave like the single-threaded version, executing each task synchronously before returning from `execute`, as shown in `WithinThreadExecutor` in Listing 6.6.

6.2.2 Execution policies

The value of decoupling submission from execution is that it lets you easily specify, and subsequently change without great difficulty, the *execution policy* for a given class of tasks. An execution policy specifies the “what, where, when, and how” of task execution, including:

```
public class WithinThreadExecutor implements Executor {  
    public void execute(Runnable r) {  
        r.run();  
    };  
}
```

LISTING 6.6. Executor that executes tasks synchronously in the calling thread.

- In what thread will tasks be executed?
- In what order should tasks be executed (FIFO, LIFO, priority order)?
- How many tasks may execute concurrently?
- How many tasks may be queued pending execution?
- If a task has to be rejected because the system is overloaded, which task should be selected as the victim, and how should the application be notified?
- What actions should be taken before or after executing a task?

Execution policies are a resource management tool, and the optimal policy depends on the available computing resources and your quality-of-service requirements. By limiting the number of concurrent tasks, you can ensure that the application does not fail due to resource exhaustion or suffer performance problems due to contention for scarce resources.³ Separating the specification of execution policy from task submission makes it practical to select an execution policy at deployment time that is matched to the available hardware.

Whenever you see code of the form:

```
new Thread(runnable).start()
```

and you think you might at some point want a more flexible execution policy, seriously consider replacing it with the use of an Executor.

6.2.3 Thread pools

A thread pool, as its name suggests, manages a homogeneous pool of worker threads. A thread pool is tightly bound to a *work queue* holding tasks waiting to be executed. Worker threads have a simple life: request the next task from the work queue, execute it, and go back to waiting for another task.

3. This is analogous to one of the roles of a transaction monitor in an enterprise application: it can throttle the rate at which transactions are allowed to proceed so as not to exhaust or overstress limited resources.

Executing tasks in pool threads has a number of advantages over the thread-per-task approach. Reusing an existing thread instead of creating a new one amortizes thread creation and teardown costs over multiple requests. As an added bonus, since the worker thread often already exists at the time the request arrives, the latency associated with thread creation does not delay task execution, thus improving responsiveness. By properly tuning the size of the thread pool, you can have enough threads to keep the processors busy while not having so many that your application runs out of memory or thrashes due to competition among threads for resources.

The class library provides a flexible thread pool implementation along with some useful predefined configurations. You can create a thread pool by calling one of the static factory methods in `Executors`:

`newFixedThreadPool`. A fixed-size thread pool creates threads as tasks are submitted, up to the maximum pool size, and then attempts to keep the pool size constant (adding new threads if a thread dies due to an unexpected `Exception`).

`newCachedThreadPool`. A cached thread pool has more flexibility to reap idle threads when the current size of the pool exceeds the demand for processing, and to add new threads when demand increases, but places no bounds on the size of the pool.

`newSingleThreadExecutor`. A single-threaded executor creates a single worker thread to process tasks, replacing it if it dies unexpectedly. Tasks are guaranteed to be processed sequentially according to the order imposed by the task queue (FIFO, LIFO, priority order).⁴

`newScheduledThreadPool`. A fixed-size thread pool that supports delayed and periodic task execution, similar to `Timer`. (See Section 6.2.5.)

The `newFixedThreadPool` and `newCachedThreadPool` factories return instances of the general-purpose `ThreadPoolExecutor`, which can also be used directly to construct more specialized executors. We discuss thread pool configuration options in depth in Chapter 8.

The web server in `TaskExecutionWebServer` uses an `Executor` with a bounded pool of worker threads. Submitting a task with `execute` adds the task to the work queue, and the worker threads repeatedly dequeue tasks from the work queue and execute them.

Switching from a thread-per-task policy to a pool-based policy has a big effect on application stability: the web server will no longer fail under heavy load.⁵

4. Single-threaded executors also provide sufficient internal synchronization to guarantee that any memory writes made by tasks are visible to subsequent tasks; this means that objects can be safely confined to the “task thread” even though that thread may be replaced with another from time to time.

5. While the server may not fail due to the creation of too many threads, if the task arrival rate exceeds the task service rate for long enough it is still possible (just harder) to run out of memory because of the growing queue of `Runnable`s awaiting execution. This can be addressed within the `Executor` framework by using a bounded work queue—see Section 8.3.2.

It also degrades more gracefully, since it does not create thousands of threads that compete for limited CPU and memory resources. And using an Executor opens the door to all sorts of additional opportunities for tuning, management, monitoring, logging, error reporting, and other possibilities that would have been far more difficult to add without a task execution framework.

6.2.4 Executor lifecycle

We've seen how to create an Executor but not how to shut one down. An Executor implementation is likely to create threads for processing tasks. But the JVM can't exit until all the (nondaemon) threads have terminated, so failing to shut down an Executor could prevent the JVM from exiting.

Because an Executor processes tasks asynchronously, at any given time the state of previously submitted tasks is not immediately obvious. Some may have completed, some may be currently running, and others may be queued awaiting execution. In shutting down an application, there is a spectrum from graceful shutdown (finish what you've started but don't accept any new work) to abrupt shutdown (turn off the power to the machine room), and various points in between. Since Executors provide a service to applications, they should be able to be shut down as well, both gracefully and abruptly, and feed back information to the application about the status of tasks that were affected by the shutdown.

To address the issue of execution service lifecycle, the `ExecutorService` interface extends `Executor`, adding a number of methods for lifecycle management (as well as some convenience methods for task submission). The lifecycle management methods of `ExecutorService` are shown in Listing 6.7.

```
public interface ExecutorService extends Executor {
    void shutdown();
    List<Runnable> shutdownNow();
    boolean isShutdown();
    boolean isTerminated();
    boolean awaitTermination(long timeout, TimeUnit unit)
        throws InterruptedException;
    // ... additional convenience methods for task submission
}
```

LISTING 6.7. Lifecycle methods in `ExecutorService`.

The lifecycle implied by `ExecutorService` has three states—*running*, *shutting down*, and *terminated*. `ExecutorServices` are initially created in the *running* state. The `shutdown` method initiates a graceful shutdown: no new tasks are accepted but previously submitted tasks are allowed to complete—including those that have not yet begun execution. The `shutdownNow` method initiates an abrupt shutdown: it attempts to cancel outstanding tasks and does not start any tasks that are queued but not begun.

Tasks submitted to an `ExecutorService` after it has been shut down are handled by the *rejected execution handler* (see Section 8.3.3), which might silently dis-

card the task or might cause `execute` to throw the unchecked `RejectedExecutionException`. Once all tasks have completed, the `ExecutorService` transitions to the *terminated* state. You can wait for an `ExecutorService` to reach the terminated state with `awaitTermination`, or poll for whether it has yet terminated with `isTerminated`. It is common to follow shutdown immediately by `awaitTermination`, creating the effect of synchronously shutting down the `ExecutorService`. (Executor shutdown and task cancellation are covered in more detail in Chapter 7.)

`LifecycleWebServer` in Listing 6.8 extends our web server with lifecycle support. It can be shut down in two ways: programmatically by calling `stop`, and through a client request by sending the web server a specially formatted HTTP request.

```
class LifecycleWebServer {
    private final ExecutorService exec = ...;

    public void start() throws IOException {
        ServerSocket socket = new ServerSocket(80);
        while (!exec.isShutdown()) {
            try {
                final Socket conn = socket.accept();
                exec.execute(new Runnable() {
                    public void run() { handleRequest(conn); }
                });
            } catch (RejectedExecutionException e) {
                if (!exec.isShutdown())
                    log("task submission rejected", e);
            }
        }
    }

    public void stop() { exec.shutdown(); }

    void handleRequest(Socket connection) {
        Request req = readRequest(connection);
        if (isShutdownRequest(req))
            stop();
        else
            dispatchRequest(req);
    }
}
```

LISTING 6.8. Web server with shutdown support.

6.2.5 Delayed and periodic tasks

The `Timer` facility manages the execution of deferred (“run this task in 100 ms”) and periodic (“run this task every 10 ms”) tasks. However, `Timer` has some drawbacks, and `ScheduledThreadPoolExecutor` should be thought of as its replacement.⁶ You can construct a `ScheduledThreadPoolExecutor` through its constructor or through the `newScheduledThreadPool` factory.

A `Timer` creates only a single thread for executing timer tasks. If a timer task takes too long to run, the timing accuracy of other `TimerTasks` can suffer. If a recurring `TimerTask` is scheduled to run every 10 ms and another `TimerTask` takes 40 ms to run, the recurring task either (depending on whether it was scheduled at fixed rate or fixed delay) gets called four times in rapid succession after the long-running task completes, or “misses” four invocations completely. Scheduled thread pools address this limitation by letting you provide multiple threads for executing deferred and periodic tasks.

Another problem with `Timer` is that it behaves poorly if a `TimerTask` throws an unchecked exception. The `Timer` thread doesn’t catch the exception, so an unchecked exception thrown from a `TimerTask` terminates the timer thread. `Timer` also doesn’t resurrect the thread in this situation; instead, it erroneously assumes the entire `Timer` was cancelled. In this case, `TimerTasks` that are already scheduled but not yet executed are never run, and new tasks cannot be scheduled. (This problem, called “thread leakage” is described in Section 7.3, along with techniques for avoiding it.)

`OutOfTime` in Listing 6.9 illustrates how a `Timer` can become confused in this manner and, as confusion loves company, how the `Timer` shares its confusion with the next hapless caller that tries to submit a `TimerTask`. You might expect the program to run for six seconds and exit, but what actually happens is that it terminates after one second with an `IllegalStateException` whose message text is “Timer already cancelled”. `ScheduledThreadPoolExecutor` deals properly with ill-behaved tasks; there is little reason to use `Timer` in Java 5.0 or later.

If you need to build your own scheduling service, you may still be able to take advantage of the library by using a `DelayQueue`, a `BlockingQueue` implementation that provides the scheduling functionality of `ScheduledThreadPoolExecutor`. A `DelayQueue` manages a collection of `Delayed` objects. A `Delayed` has a delay time associated with it: `DelayQueue` lets you take an element only if its delay has expired. Objects are returned from a `DelayQueue` ordered by the time associated with their delay.

6.3 Finding exploitable parallelism

The `Executor` framework makes it easy to specify an execution policy, but in order to use an `Executor`, you have to be able to describe your task as a `Runnable`. In most server applications, there is an obvious task boundary: a single client request. But sometimes good task boundaries are not quite so obvious, as

6. `Timer` does have support for scheduling based on absolute, not relative time, so that tasks can be sensitive to changes in the system clock; `ScheduledThreadPoolExecutor` supports only relative time.

```
public class OutOfTime {
    public static void main(String[] args) throws Exception {
        Timer timer = new Timer();
        timer.schedule(new ThrowTask(), 1);
        SECONDS.sleep(1);
        timer.schedule(new ThrowTask(), 1);
        SECONDS.sleep(5);
    }

    static class ThrowTask extends TimerTask {
        public void run() { throw new RuntimeException(); }
    }
}
```



LISTING 6.9. Class illustrating confusing Timer behavior.

in many desktop applications. There may also be exploitable parallelism within a single client request in server applications, as is sometimes the case in database servers. (For a further discussion of the competing design forces in choosing task boundaries, see [CPJ 4.4.1.1].)

In this section we develop several versions of a component that admit varying degrees of concurrency. Our sample component is the page-rendering portion of a browser application, which takes a page of HTML and renders it into an image buffer. To keep it simple, we assume that the HTML consists only of marked up text interspersed with image elements with pre-specified dimensions and URLs.

6.3.1 Example: sequential page renderer

The simplest approach is to process the HTML document sequentially. As text markup is encountered, render it into the image buffer; as image references are encountered, fetch the image over the network and draw it into the image buffer as well. This is easy to implement and requires touching each element of the input only once (it doesn't even require buffering the document), but is likely to annoy the user, who may have to wait a long time before all the text is rendered.

A less annoying but still sequential approach involves rendering the text elements first, leaving rectangular placeholders for the images, and after completing the initial pass on the document, going back and downloading the images and drawing them into the associated placeholder. This approach is shown in `SingleThreadRenderer` in Listing 6.10.

Downloading an image mostly involves waiting for I/O to complete, and during this time the CPU does little work. So the sequential approach may underutilize the CPU, and also makes the user wait longer than necessary to see the finished page. We can achieve better utilization and responsiveness by breaking the problem into independent tasks that can execute concurrently.

```
public class SingleThreadRenderer {  
    void renderPage(CharSequence source) {  
        renderText(source);  
        List<ImageData> imageData = new ArrayList<ImageData>();  
        for (ImageInfo imageInfo : scanForImageInfo(source))  
            imageData.add(imageInfo.downloadImage());  
        for (ImageData data : imageData)  
            renderImage(data);  
    }  
}
```



LISTING 6.10. Rendering page elements sequentially.

6.3.2 Result-bearing tasks: `Callable` and `Future`

The `Executor` framework uses `Runnable` as its basic task representation. `Runnable` is a fairly limiting abstraction; run cannot return a value or throw checked exceptions, although it can have side effects such as writing to a log file or placing a result in a shared data structure.

Many tasks are effectively deferred computations—executing a database query, fetching a resource over the network, or computing a complicated function. For these types of tasks, `Callable` is a better abstraction: it expects that the main entry point, `call`, will return a value and anticipates that it might throw an exception.⁷ `Executors` includes several utility methods for wrapping other types of tasks, including `Runnable` and `java.security.PrivilegedAction`, with a `Callable`.

`Runnable` and `Callable` describe abstract computational tasks. Tasks are usually finite: they have a clear starting point and they eventually terminate. The lifecycle of a task executed by an `Executor` has four phases: *created*, *submitted*, *started*, and *completed*. Since tasks can take a long time to run, we also want to be able to cancel a task. In the `Executor` framework, tasks that have been submitted but not yet started can always be cancelled, and tasks that have started can sometimes be cancelled if they are responsive to interruption. Cancelling a task that has already completed has no effect. (Cancellation is covered in greater detail in Chapter 7.)

`Future` represents the lifecycle of a task and provides methods to test whether the task has completed or been cancelled, retrieve its result, and cancel the task. `Callable` and `Future` are shown in Listing 6.11. Implicit in the specification of `Future` is that task lifecycle can only move forwards, not backwards—just like the `ExecutorService` lifecycle. Once a task is completed, it stays in that state forever.

The behavior of `get` varies depending on the task state (not yet started, running, completed). It returns immediately or throws an `Exception` if the task has already completed, but if not it blocks until the task completes. If the task completes by throwing an exception, `get` rethrows it wrapped in an `Execution-`

7. To express a non-value-returning task with `Callable`, use `Callable<Void>`.

```
public interface Callable<V> {
    V call() throws Exception;
}

public interface Future<V> {
    boolean cancel(boolean mayInterruptIfRunning);
    boolean isCancelled();
    boolean isDone();
    V get() throws InterruptedException, ExecutionException,
        CancellationException;
    V get(long timeout, TimeUnit unit)
        throws InterruptedException, ExecutionException,
        CancellationException, TimeoutException;
}
```

LISTING 6.11. Callable and Future interfaces.

Exception; if it was cancelled, `get` throws `CancellationException`. If `get` throws `ExecutionException`, the underlying exception can be retrieved with `getCause`.

There are several ways to create a `Future` to describe a task. The `submit` methods in `ExecutorService` all return a `Future`, so that you can submit a `Runnable` or a `Callable` to an executor and get back a `Future` that can be used to retrieve the result or cancel the task. You can also explicitly instantiate a `FutureTask` for a given `Runnable` or `Callable`. (Because `FutureTask` implements `Runnable`, it can be submitted to an `Executor` for execution or executed directly by calling its `run` method.)

As of Java 6, `ExecutorService` implementations can override `newTaskFor` in `AbstractExecutorService` to control instantiation of the `Future` corresponding to a submitted `Callable` or `Runnable`. The default implementation just creates a new `FutureTask`, as shown in Listing 6.12.

```
protected <T> RunnableFuture<T> newTaskFor(Callable<T> task) {
    return new FutureTask<T>(task);
}
```

LISTING 6.12. Default implementation of `newTaskFor` in `ThreadPoolExecutor`.

Submitting a `Runnable` or `Callable` to an `Executor` constitutes a safe publication (see Section 3.5) of the `Runnable` or `Callable` from the submitting thread to the thread that will eventually execute the task. Similarly, setting the result value for a `Future` constitutes a safe publication of the result from the thread in which it was computed to any thread that retrieves it via `get`.

6.3.3 Example: page renderer with Future

As a first step towards making the page renderer more concurrent, let's divide it into two tasks, one that renders the text and one that downloads all the images. (Because one task is largely CPU-bound and the other is largely I/O-bound, this approach may yield improvements even on single-CPU systems.)

`Callable` and `Future` can help us express the interaction between these cooperating tasks. In `FutureRenderer` in Listing 6.13, we create a `Callable` to download all the images, and submit it to an `ExecutorService`. This returns a `Future` describing the task's execution; when the main task gets to the point where it needs the images, it waits for the result by calling `Future.get`. If we're lucky, the results will already be ready by the time we ask; otherwise, at least we got a head start on downloading the images.

The state-dependent nature of `get` means that the caller need not be aware of the state of the task, and the safe publication properties of task submission and result retrieval make this approach thread-safe. The exception handling code surrounding `Future.get` deals with two possible problems: that the task encountered an `Exception`, or the thread calling `get` was interrupted before the results were available. (See Sections 5.5.2 and 5.4.)

`FutureRenderer` allows the text to be rendered concurrently with downloading the image data. When all the images are downloaded, they are rendered onto the page. This is an improvement in that the user sees a result quickly and it exploits some parallelism, but we can do considerably better. There is no need for users to wait for *all* the images to be downloaded; they would probably prefer to see individual images drawn as they become available.

6.3.4 Limitations of parallelizing heterogeneous tasks

In the last example, we tried to execute two different types of tasks in parallel—downloading the images and rendering the page. But obtaining significant performance improvements by trying to parallelize sequential heterogeneous tasks can be tricky.

Two people can divide the work of cleaning the dinner dishes fairly effectively: one person washes while the other dries. However, assigning a different type of task to each worker does not scale well; if several more people show up, it is not obvious how they can help without getting in the way or significantly restructuring the division of labor. Without finding finer-grained parallelism among similar tasks, this approach will yield diminishing returns.

A further problem with dividing heterogeneous tasks among multiple workers is that the tasks may have disparate sizes. If you divide tasks *A* and *B* between two workers but *A* takes ten times as long as *B*, you've only speeded up the total process by 9%. Finally, dividing a task among multiple workers always involves some amount of coordination overhead; for the division to be worthwhile, this overhead must be more than compensated by productivity improvements due to parallelism.

`FutureRenderer` uses two tasks: one for rendering text and one for downloading the images. If rendering the text is much faster than downloading the images,

```

public class FutureRenderer {
    private final ExecutorService executor = ...;

    void renderPage(CharSequence source) {
        final List<ImageInfo> imageInfos = scanForImageInfo(source);
        Callable<List<ImageData>> task =
            new Callable<List<ImageData>>() {
                public List<ImageData> call() {
                    List<ImageData> result
                        = new ArrayList<ImageData>();
                    for (ImageInfo imageInfo : imageInfos)
                        result.add(imageInfo.downloadImage());
                    return result;
                }
            };

        Future<List<ImageData>> future = executor.submit(task);
        renderText(source);

        try {
            List<ImageData> imageData = future.get();
            for (ImageData data : imageData)
                renderImage(data);
        } catch (InterruptedException e) {
            // Re-assert the thread's interrupted status
            Thread.currentThread().interrupt();
            // We don't need the result, so cancel the task too
            future.cancel(true);
        } catch (ExecutionException e) {
            throw launderThrowable(e.getCause());
        }
    }
}

```



LISTING 6.13. Waiting for image download with Future.

as is entirely possible, the resulting performance is not much different from the sequential version, but the code is a lot more complicated. And the best we can do with two threads is speed things up by a factor of two. Thus, trying to increase concurrency by parallelizing heterogeneous activities can be a lot of work, and there is a limit to how much additional concurrency you can get out of it. (See Sections 11.4.2 and 11.4.3 for another example of the same phenomenon.)

The real performance payoff of dividing a program's workload into tasks comes when there are a large number of independent, *homogeneous* tasks that can be processed concurrently.

6.3.5 CompletionService: Executor meets BlockingQueue

If you have a batch of computations to submit to an `Executor` and you want to retrieve their results as they become available, you could retain the `Future` associated with each task and repeatedly poll for completion by calling `get` with a timeout of zero. This is possible, but tedious. Fortunately there is a better way: a *completion service*.

`CompletionService` combines the functionality of an `Executor` and a `BlockingQueue`. You can submit `Callable` tasks to it for execution and use the queue-like methods `take` and `poll` to retrieve completed results, packaged as `Futures`, as they become available. `ExecutorCompletionService` implements `CompletionService`, delegating the computation to an `Executor`.

The implementation of `ExecutorCompletionService` is quite straightforward. The constructor creates a `BlockingQueue` to hold the completed results. `FutureTask` has a `done` method that is called when the computation completes. When a task is submitted, it is wrapped with a `QueueingFuture`, a subclass of `FutureTask` that overrides `done` to place the result on the `BlockingQueue`, as shown in Listing 6.14. The `take` and `poll` methods delegate to the `BlockingQueue`, blocking if results are not yet available.

```
private class QueueingFuture<V> extends FutureTask<V> {
    QueueingFuture(Callable<V> c) { super(c); }
    QueueingFuture(Runnable t, V r) { super(t, r); }

    protected void done() {
        completionQueue.add(this);
    }
}
```

LISTING 6.14. `QueueingFuture` class used by `ExecutorCompletionService`.

6.3.6 Example: page renderer with `CompletionService`

We can use a `CompletionService` to improve the performance of the page renderer in two ways: shorter total runtime and improved responsiveness. We can create a separate task for downloading *each* image and execute them in a thread pool, turning the sequential download into a parallel one: this reduces the amount of time to download all the images. And by fetching results from the `CompletionService` and rendering each image as soon as it is available, we can give the user a more dynamic and responsive user interface. This implementation is shown in `Renderer` in Listing 6.15.

```
public class Renderer {
    private final ExecutorService executor;

    Renderer(ExecutorService executor) { this.executor = executor; }

    void renderPage(CharSequence source) {
        List<ImageInfo> info = scanForImageInfo(source);
        CompletionService<ImageData> completionService =
            new ExecutorCompletionService<ImageData>(executor);
        for (final ImageInfo imageInfo : info)
            completionService.submit(new Callable<ImageData>() {
                public ImageData call() {
                    return imageInfo.downloadImage();
                }
            });

        renderText(source);

        try {
            for (int t = 0, n = info.size(); t < n; t++) {
                Future<ImageData> f = completionService.take();
                ImageData imageData = f.get();
                renderImage(imageData);
            }
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        } catch (ExecutionException e) {
            throw launderThrowable(e.getCause());
        }
    }
}
```

LISTING 6.15. Using `CompletionService` to render page elements as they become available.

Multiple `ExecutorCompletionServices` can share a single `Executor`, so it is

perfectly sensible to create an `ExecutorCompletionService` that is private to a particular computation while sharing a common `Executor`. When used in this way, a `CompletionService` acts as a handle for a batch of computations in much the same way that a `Future` acts as a handle for a single computation. By remembering how many tasks were submitted to the `CompletionService` and counting how many completed results are retrieved, you can know when all the results for a given batch have been retrieved, even if you use a shared `Executor`.

6.3.7 Placing time limits on tasks

Sometimes, if an activity does not complete within a certain amount of time, the result is no longer needed and the activity can be abandoned. For example, a web application may fetch its advertisements from an external ad server, but if the ad is not available within two seconds, it instead displays a default advertisement so that ad unavailability does not undermine the site's responsiveness requirements. Similarly, a portal site may fetch data in parallel from multiple data sources, but may be willing to wait only a certain amount of time for data to be available before rendering the page without it.

The primary challenge in executing tasks within a time budget is making sure that you don't wait longer than the time budget to get an answer or find out that one is not forthcoming. The timed version of `Future.get` supports this requirement: it returns as soon as the result is ready, but throws `TimeoutException` if the result is not ready within the timeout period.

A secondary problem when using timed tasks is to stop them when they run out of time, so they do not waste computing resources by continuing to compute a result that will not be used. This can be accomplished by having the task strictly manage its own time budget and abort if it runs out of time, or by cancelling the task if the timeout expires. Again, `Future` can help; if a timed `get` completes with a `TimeoutException`, you can cancel the task through the `Future`. If the task is written to be cancellable (see Chapter 7), it can be terminated early so as not to consume excessive resources. This technique is used in Listings 6.13 and 6.16.

Listing 6.16 shows a typical application of a timed `Future.get`. It generates a composite web page that contains the requested content plus an advertisement fetched from an ad server. It submits the ad-fetching task to an executor, computes the rest of the page content, and then waits for the ad until its time budget runs out.⁸ If the `get` times out, it cancels⁹ the ad-fetching task and uses a default advertisement instead.

6.3.8 Example: a travel reservations portal

The time-budgeting approach in the previous section can be easily generalized to an arbitrary number of tasks. Consider a travel reservation portal: the user en-

8. The timeout passed to `get` is computed by subtracting the current time from the deadline; this may in fact yield a negative number, but all the timed methods in `java.util.concurrent` treat negative timeouts as zero, so no extra code is needed to deal with this case.

9. The true parameter to `Future.cancel` means that the task thread can be interrupted if the task is currently running; see Chapter 7.

```

Page renderPageWithAd() throws InterruptedException {
    long endNanos = System.nanoTime() + TIME_BUDGET;
    Future<Ad> f = exec.submit(new FetchAdTask());
    // Render the page while waiting for the ad
    Page page = renderPageBody();
    Ad ad;
    try {
        // Only wait for the remaining time budget
        long timeLeft = endNanos - System.nanoTime();
        ad = f.get(timeLeft, NANOSECONDS);
    } catch (ExecutionException e) {
        ad = DEFAULT_AD;
    } catch (TimeoutException e) {
        ad = DEFAULT_AD;
        f.cancel(true);
    }
    page.setAd(ad);
    return page;
}

```

LISTING 6.16. Fetching an advertisement with a time budget.

ters travel dates and requirements and the portal fetches and displays bids from a number of airlines, hotels or car rental companies. Depending on the company, fetching a bid might involve invoking a web service, consulting a database, performing an EDI transaction, or some other mechanism. Rather than have the response time for the page be driven by the slowest response, it may be preferable to present only the information available within a given time budget. For providers that do not respond in time, the page could either omit them completely or display a placeholder such as “Did not hear from Air Java in time.”

Fetching a bid from one company is independent of fetching bids from another, so fetching a single bid is a sensible task boundary that allows bid retrieval to proceed concurrently. It would be easy enough to create n tasks, submit them to a thread pool, retain the Futures, and use a timed get to fetch each result sequentially via its Future, but there is an even easier way—`invokeAll`.

Listing 6.17 uses the timed version of `invokeAll` to submit multiple tasks to an `ExecutorService` and retrieve the results. The `invokeAll` method takes a collection of tasks and returns a collection of Futures. The two collections have identical structures; `invokeAll` adds the Futures to the returned collection in the order imposed by the task collection’s iterator, thus allowing the caller to associate a Future with the `Callable` it represents. The timed version of `invokeAll` will return when all the tasks have completed, the calling thread is interrupted, or the timeout expires. Any tasks that are not complete when the timeout expires are cancelled. On return from `invokeAll`, each task will have either completed normally or been cancelled; the client code can call `get` or `isCancelled` to find

out which.

Summary

Structuring applications around the execution of *tasks* can simplify development and facilitate concurrency. The `Executor` framework permits you to decouple task submission from execution policy and supports a rich variety of execution policies; whenever you find yourself creating threads to perform tasks, consider using an `Executor` instead. To maximize the benefit of decomposing an application into tasks, you must identify sensible task boundaries. In some applications, the obvious task boundaries work well, whereas in others some analysis may be required to uncover finer-grained exploitable parallelism.

```
private class QuoteTask implements Callable<TravelQuote> {
    private final TravelCompany company;
    private final TravelInfo travelInfo;
    ...
    public TravelQuote call() throws Exception {
        return company.solicitQuote(travelInfo);
    }
}

public List<TravelQuote> getRankedTravelQuotes(
    TravelInfo travelInfo, Set<TravelCompany> companies,
    Comparator<TravelQuote> ranking, long time, TimeUnit unit)
    throws InterruptedException {
    List<QuoteTask> tasks = new ArrayList<QuoteTask>();
    for (TravelCompany company : companies)
        tasks.add(new QuoteTask(company, travelInfo));

    List<Future<TravelQuote>> futures =
        exec.invokeAll(tasks, time, unit);

    List<TravelQuote> quotes =
        new ArrayList<TravelQuote>(tasks.size());
    Iterator<QuoteTask> taskIter = tasks.iterator();
    for (Future<TravelQuote> f : futures) {
        QuoteTask task = taskIter.next();
        try {
            quotes.add(f.get());
        } catch (ExecutionException e) {
            quotes.add(task.getFailureQuote(e.getCause()));
        } catch (CancellationException e) {
            quotes.add(task.getTimeoutQuote(e));
        }
    }

    Collections.sort(quotes, ranking);
    return quotes;
}
```

LISTING 6.17. Requesting travel quotes under a time budget.

This page intentionally left blank

Index

Symbols

64-bit operations

nonatomic nature of; 36

A

ABA problem; 336

abnormal thread termination

handling; 161–163

abort saturation policy; 174

See also lifecycle; termination;

abrupt shutdown

limitations; 158–161

triggers for; 164

vs. graceful shutdown; 153

AbstractExecutorService

task representation use; 126

abstractions

See models/modeling; representation;

AbstractQueuedSynchronizer

See AQS framework;

access

See also encapsulation; sharing; visibility;

exclusive

and concurrent collections; 86

integrity

nonblocking algorithm use; 319

mutable state

importance of coordinating; 110

remote resource

as long-running GUI task; 195

serialized

WorkerThread example; 227_{li}

vs. object serialization; 27_{fn}

visibility role in; 33

AccessControlContext

custom thread factory handling; 177

acquisition of locks

See locks, acquisition;

action(s)

See also compound actions; condition, predicate; control flow; task(s);

barrier; 99

JMM specification; 339–342

listener; 195–197

activity(s)

See also task(s);

cancellation; 135, 135–150

tasks as representation of; 113

ad-hoc thread confinement; 43

See also confinement;

algorithm(s)

See also design patterns; idioms; representation;

comparing performance; 263–264

design role of representation; 104

lock-free; 329

Michael-Scott nonblocking queue; 332

nonblocking; 319, 329, 329–336

backoff importance for; 231_{fn}

synchronization; 319–336

SynchronousQueue; 174_{fn}

parallel iterative

barrier use in; 99

recursive

parallelizing; 181–188

Treiber's

nonblocking stack; 331_{li}

work stealing

dequeues and; 92

alien method; 40

See also untrusted code behavior;

deadlock risks posed by; 211

publication risks; 40

allocation

- advantages vs. synchronization; 242
- immutable objects and; 48_{fn}
- object pool use
 - disadvantages of; 241
- scalability advantages; 263

Amdahl's law; 225–229

- See also* concurrent/concurrency;
 - performance; resource(s);
 - throughput; utilization;
- lock scope reduction advantage; 234
- qualitative application of; 227

analysis

- See also* instrumentation; measure-
ment; static analysis tools;
- deadlock
 - thread dump use; 216–217
- escape; 230
- for exploitable parallelism; 123–133
- lock contention
 - thread dump use; 240
- performance; 221–245

annotations; 353–354

- See also* documentation;
- for concurrency documentation; 6
- @GuardedBy; 28, 354
 - synchronization policy docu-
mentation use; 75
- @Immutable; 353
- @NotThreadSafe; 353
- @ThreadSafe; 353

AOP (aspect-oriented programming)

- in testing; 259, 273

application(s)

- See also* frameworks(s); service(s);
tools;
- scoped objects
 - thread safety concerns; 10
- frameworks, and ThreadLocal; 46
- GUI; 189–202
 - thread safety concerns; 10–11
- parallelizing
 - task decomposition; 113
- shutdown
 - and task cancellation; 136

AQS (AbstractQueuedSynchronizer)

- framework**; 308, 311–317
- exit protocol use; 306
- FutureTask implementation
 - piggybacking use; 342

ArrayBlockingQueue; 89

- as bounded buffer example; 292
- performance advantages over
BoundedBuffer; 263

ArrayDeque; 92**arrays**

- See also* collections; data struc-
ture(s);
- atomic variable; 325

asymmetric two-party tasks

- Exchanger management of; 101

asynchrony/asynchronous

- events, handling; 4
- I/O, and non-interruptable block-
ing; 148
- sequentiality vs.; 2
- tasks
 - execution, Executor framework
use; 117
 - FutureTask handling; 95–98

atomic variables; 319–336

- and lock contention; 239–240
- classes; 324–329
- locking vs.; 326–329
- strategies for use; 34
- volatile variables vs.; 39, 324–326

atomic/atomicity; 22

- See also* invariant(s); synchroniza-
tion; visibility;
- 64-bit operations
 - nonatomic nature of; 36
- and compound actions; 22–23
- and multivariable invariants; 57,
67–68
- and open call restructuring; 213
- and service shutdown; 153
- and state transition constraints; 56
- caching issues; 24–25
- client-side locking support for; 80
- field updaters; 335–336
- immutable object use for; 48
- in cache implementation design; 106
- intrinsic lock enforcement of; 25–26
- loss
 - risk of lock scope reduction; 234
- Map operations; 86
- put-if-absent; 71–72
- statistics gathering hooks use; 179
- thread-safety issues
 - in servlets with state; 19–23

AtomicBoolean; 325
AtomicInteger; 324
 nonblocking algorithm use; 319
 random number generator using;
 327_{li}
AtomicLong; 325
AtomicReference; 325
 nonblocking algorithm use; 319
 safe publication use; 52
AtomicReferenceFieldUpdater; 335
audit(ing)
 See also instrumentation;
audit(ing) tools; 28_{fi}
AWT (Abstract Window Toolkit)
 See also GUI;
 thread use; 9
 safety concerns and; 10–11

B

backoff
 and nonblocking algorithms; 231_{fi}
barging; 283
 See also fairness; ordering; synchro-
 nization;
 and read-write locks; 287
 performance advantages of; 284
barrier(s); 99, 99–101
 See also latch(es); semaphores; syn-
 chronizers;
 -based timer; 260–261
 action; 99
 memory; 230, 338
 point; 99
behavior
 See also activities; task(s);
bias
 See testing, pitfalls;
bibliography; 355–357
binary latch; 304
 AQS-based; 313–314
binary semaphore
 mutex use; 99
Bloch, Joshua
 (bibliographic reference); 69
block(ing); 92
 bounded collections
 semaphore management of; 99
 testing; 248
 context switching impact of; 230
 interruptible methods and; 92–94
 interruption handling methods; 138

 methods
 and interruption; 143
 non-interruptable; 147–150
 operations
 testing; 250–252
 thread pool size impact; 170
 queues; 87–94
 See also Semaphore;
 and thread pool management;
 173
 cancellation, problems; 138
 cancellation, solutions; 140
 Executor functionality com-
 bined with; 129
 producer-consumer pattern and;
 87–92
 spin-waiting; 232
 state-dependent actions; 291–308
 and polling; 295–296
 and sleeping; 295–296
 condition queues; 296–308
 structure; 292_{li}
 threads, costs of; 232
 waits
 timed vs. unbounded; 170
BlockingQueue; 84–85
 and state-based preconditions; 57
 safe publication use; 52
 thread pool use of; 173
bound(ed)
 See also constraints; encapsulation;
 blocking collections
 semaphore management of; 99
 buffers
 blocking operations; 292
 scalability testing; 261
 size determination; 261
 queues
 and producer-consumer pattern;
 88
 saturation policies; 174–175
 thread pool use; 172
 thread pool use of; 173
 resource; 221
boundaries
 See also encapsulation;
 task; 113
 analysis for parallelism; 123–133
broken multi-threaded programs
 strategies for fixing; 16

BrokenBarrierException

parallel iterative algorithm use; 99

buffer(s)

See also cache/caching;

bounded

blocking state-dependent operations with; 292

scalability testing; 261

size determination; 261

BoundedBuffer example; 249_{li}

condition queue use; 297

test case development for; 248

BoundedBufferTest example; 250_{li}

capacities

comparison testing; 261–263

testing; 248

bug pattern(s); 271, 271

See also debugging; design patterns; testing;

detector; 271

busy-waiting; 295

See also spin-waiting;

C**cache/caching**

See also performance;

atomicity issues; 24–25

flushing

and memory barriers; 230

implementation issues

atomic/atomicity; 106

safety; 104

misses

as cost of context switching; 229

result

building; 101–109

Callable; 126_{li}

FutureTask use; 95

results handling capabilities; 125

callbacks

testing use; 257–259

caller-runs saturation policy; 174**cancellation; 135–150**

See also interruption; lifecycle; shutdown;

activity; 135

as form of completion; 95

Future use; 145–147

interruptible lock acquisition; 279–281

interruption relationship to; 138

long-running GUI tasks; 197–198

non-standard

encapsulation of; 148–150

reasons and strategies; 147–150

points; 140

policy; 136

and thread interruption policy; 141

interruption advantages as implementation strategy; 140

reasons for; 136

shutdown and; 135–166

task

Executor handling; 125

in timed task handling; 131

timed locks use; 279

CancellationException

Callable handling; 98

CAS (compare-and-swap) instructions;

321–324

See also atomic/atomicity, variables;

Java class support in Java 5.0; 324

lock-free algorithm use; 329

nonblocking algorithm use; 319, 329

cascading effects

of thread safety requirements; 28

cellular automata

barrier use for computation of; 101

check-then-act operation

See also compound actions;

as race condition cause; 21

atomic variable handling; 325

compound action

in collection operations; 79

multivariable invariant issues; 67–68

service shutdown issue; 153

checkpoint

state

shutdown issues; 158

checksums

safety testing use; 253

class(es)

as instance confinement context; 59

extension

strategies and risks; 71

with helper classes; 72–73

synchronized wrapper

client-side locking support; 73

thread-safe

and object composition; 55–78

cleanup

See also lifecycle;
and interruption handling
protecting data integrity; 142
in end-of-lifecycle processing; 135
JVM shutdown hooks use for; 164

client(s)

See also server;
requests
as natural task boundary; 113

client-side locking; 72–73, 73

See also lock(ing);
and compound actions; 79–82
and condition queues; 306
class extension relationship to; 73
stream class management; 150_{fn}

coarsening

See also lock(ing);
lock; 231, 235_{fn}, 286

code review

as quality assurance strategy; 271

collections

See also hashtables; lists; set(s);
bounded blocking
semaphore management of; 99
concurrent; 84–98
building block; 79–110
copying
as alternative to locking; 83
lock striping use; 237
synchronized; 79–84
concurrent collections vs.; 84

Collections.synchronizedList

safe publication use; 52

Collections.synchronizedXxx

synchronized collection creation; 79

communication

mechanisms for; 1

compare-and-swap (CAS) instructions

See CAS;

comparison

priority-ordered queue use; 89

compilation

dynamic
and performance testing; 267–
268
timing and ordering alterations
thread safety risks; 7

completion; 95

See also lifecycle;
notification

of long-running GUI task; 198
service

Future; 129

task

measuring service time variance;
264–266
volatile variable use with; 39

CompletionService

in page rendering example; 129

composition; 73

See also delegation; encapsulation;
as robust functionality extension
mechanism; 73
of objects; 55–78

compound actions; 22

See also atomic/atomicity; concurr-
ent/concurrency, collec-
tions; race conditions;
atomicity handling of; 22–23
concurrency design rules role; 110
concurrent collection support for; 84
examples of
See check-then-act operation;
iteration; navigation; put-
if-absent operation; read-
modify-write; remove-if-
equal operation; replace-if-
equal operation;
in cache implementation; 106
in synchronized collection class use
mechanisms for handling; 79–82
synchronization requirements; 29

computation

compute-intensive code
impact on locking behavior; 34
thread pool size impact; 170
deferred
design issues; 125
thread-local
and performance testing; 268

Concurrent Programming in Java; 42,

57, 59, 87, 94, 95, 98, 99, 101,
124, 201, 211, 279, 282, 304

concurrent/concurrency

See also parallelizing/parallelism;
safety; synchroniza-
tion/synchronized;
and synchronized collections; 84
and task independence; 113
annotations; 353–354
brief history; 1–2

- building blocks; 79–110
- cache implementation issues; 103
- collections; 84–98
- ConcurrentHashMap** locking strategy
 - advantages; 85
- debugging
 - costs vs. performance optimization value; 224
- design rules; 110
- errors
 - See* deadlock; livelock; race conditions; starvation;
- fine-grained
 - and thread-safe data models; 201
- modifying
 - synchronized collection problems with; 82
- object pool disadvantages; 241
- poor; 30
- prevention
 - See also* single-threaded;
 - single-threaded executor use; 172, 177–178
- read-write lock advantages; 286–289
- testing; 247–274
- ConcurrentHashMap**; 84–86
 - performance advantages of; 242
- ConcurrentLinkedDeque**; 92
- ConcurrentLinkedQueue**; 84–85
 - algorithm; 319–336
 - reflection use; 335
 - safe publication use; 52
- ConcurrentMap**; 84, 87_{li}
 - safe publication use; 52
- ConcurrentModificationException**
 - avoiding; 85
 - fail-fast iterators use; 82–83
- ConcurrentSkipListMap**; 85
- ConcurrentSkipListSet**; 85
- Condition**; 307_{li}
 - explicit condition object use; 306
 - intrinsic condition queues vs.
 - performance considerations; 308
- condition**
 - predicate; 299, 299–300
 - lock and condition variable relationship; 308
 - queues; 297
 - See also* synchronizers;
 - AQS support for; 312
 - blocking state-dependent operations use; 296–308
 - explicit; 306–308
 - intrinsic; 297
 - intrinsic, disadvantages of; 306
 - using; 298
- variables
 - explicit; 306–308
- waits
 - and condition predicate; 299
 - canonical form; 301_{li}
 - interruptible, as feature of **Condition**; 307
 - uninterruptible, as feature of **Condition**; 307
 - waking up from, condition queue handling; 300–301
- conditional**
 - See also* blocking/blocks;
 - notification; 303
 - as optimization; 303
 - subclassing safety issues; 304
 - use; 304_{li}
 - read-modify-writer operations
 - atomic variable support for; 325
- configuration**
 - of **ThreadPoolExecutor**; 171–179
 - thread creation
 - and thread factories; 175
 - thread pool
 - post-construction manipulation; 177–179
- confinement**
 - See also* encapsulation; single-thread(ed);
 - instance; 59, 58–60
 - stack; 44, 44–45
 - thread; 42, 42–46
 - ad-hoc; 43
 - and execution policy; 167
 - in **Swing**; 191–192
 - role, synchronization policy specification; 56
 - serial; 90, 90–92
 - single-threaded GUI framework use; 190
 - ThreadLocal**; 45–46
- Connection**
 - thread confinement use; 43
 - ThreadLocal** variable use with; 45

consistent/consistency

- copy timeliness vs.
 - as design tradeoff; 62
- data view timeliness vs.
 - as design tradeoff; 66, 70
- lock ordering
 - and deadlock avoidance; 206
- weakly consistent iterators; 85

constraints

- See also* invariant(s); post-conditions;
 - pre-conditions;
- state transition; 56
- thread creation
 - importance of; 116

construction/constructors

- See also* lifecycle;
- object
 - publication risks; 41–42
 - thread handling issues; 41–42
- partial
 - unsafe publication influence; 50
- private constructor capture idiom;
 - 69_{fn}
- starting thread from
 - as concurrency bug pattern; 272
- ThreadPoolExecutor; 172_{ii}
- post-construction customization;
 - 177

consumers

- See also* blocking, queues; producer-consumer pattern;
- blocking queues use; 88
- producer-consumer pattern
 - blocking queues and; 87–92

containers

- See also* collections;
- blocking queues as; 94
- scoped
 - thread safety concerns; 10

contention/contented

- as performance inhibiting factor; 263
- intrinsic locks vs. ReentrantLock
 - performance considerations;
 - 282–286
- lock
 - costs of; 320
 - measurement; 240–241
 - reduction impact; 211
 - reduction, strategies; 232–242
 - scalability impact; 232
 - signal method reduction in; 308

- locking vs. atomic variables; 328
- resource

- and task execution policy; 119
- deque advantages; 92

scalability under

- as AQS advantage; 311

scope

- atomic variable limitation of; 324

synchronization; 230**thread**

- collision detection help with; 321
- latches help with; 95
- throughput impact; 228
- unrealistic degrees of
 - as performance testing pitfall;
 - 268–269

context switching; 229

- See also* performance;
- as cost of thread use; 229–230
- condition queues advantages; 297
- cost(s); 8
- message logging
 - reduction strategies; 243–244
- performance impact of; 221
- reduction; 243–244
- signal method reduction in; 308
- throughput impact; 228

control flow

- See also* event(s); lifecycle; MVC
 - (model-view-controller) pattern;
- coordination
 - in producer-consumer pattern;
 - 94
- event handling
 - model-view objects; 195_{fg}
 - simple; 194_{fg}
- latch characteristics; 94
- model-view-controller pattern
 - and inconsistent lock ordering;
 - 190
 - vehicle tracking example; 61

convenience

- See also* responsiveness;
- as concurrency motivation; 2

conventions

- annotations
 - concurrency documentation; 6
- Java monitor pattern; 61

cooperation/cooperating

- See also* concurrent/concurrency; synchronization;
- end-of-lifecycle mechanisms
 - interruption as; 93, 135
- model, view, and controller objects
 - in GUI applications
 - inconsistent lock ordering; 190
- objects
 - deadlock, lock-ordering; 212_{li}
 - deadlock, possibilities; 211
 - livelock possibilities; 218
- thread
 - concurrency mechanisms for; 79

coordination

- See also* synchronization/synchronized;
- control flow
 - producer-consumer pattern, blocking queues use; 94
- in multithreaded environments
 - performance impact of; 221
- mutable state access
 - importance of; 110

copying

- collections
 - as alternative to locking; 83
- data
 - thread safety consequences; 62

CopyOnWriteArrayList; 84, 86–87

- safe publication use; 52
- versioned data model use
 - in GUI applications; 201

CopyOnWriteArraySet

- safe publication use; 52
- synchronized Set replacement; 86

core pool size parameter

- thread creation impact; 171, 172_{fn}

correctly synchronized program; 341**correctness; 17**

- See also* safety;
- testing; 248–260
 - goals; 247
- thread safety defined in terms of; 17

corruption

- See also* atomic/atomicity; encapsulation; safety; state;
- data
 - and interruption handling; 142
 - causes, stale data; 35

cost(s)

- See also* guidelines; performance; safety; strategies; tradeoffs;
- thread; 229–232
 - context switching; 8
 - locality loss; 8
- tradeoffs
 - in performance optimization
 - strategies; 223

CountDownLatch; 95

- AQS use; 315–316
- puzzle-solving framework use; 184
- TestHarness example use; 96

counting semaphores; 98

- See also* Semaphore;
- permits, thread relationships; 248
- SemaphoreOnLock example; 310_{li}

coupling

- See also* dependencies;
- behavior
 - blocking queue handling; 89
- implicit
 - between tasks and execution
 - policies; 167–170

CPU utilization

- See also* performance;
- and sequential execution; 124
- condition queues advantages; 297
- impact on performance testing; 261
- monitoring; 240–241
- optimization
 - as multithreading goal; 222
- spin-waiting impact on; 295

creation

- See also* copying; design; policy(s); representation;
- atomic compound actions; 80
- class
 - existing thread-safe class reuse
 - advantages over; 71
- collection copy
 - as immutable object strategy; 86
- of immutable objects; 48
- of state-dependent methods; 57
- synchronizer; 94
- thread; 171–172
 - explicitly, for tasks; 115
 - thread factory use; 175–177
 - unbounded, disadvantages; 116
- thread pools; 120
- wrappers

during memoization; 103

customization

thread configuration

ThreadFactory use; 175

thread pool configuration

post-construction; 177–179

CyclicBarrier; 99

parallel iterative algorithm use; 102_{li}

testing use; 255_{li}, 260_{li}

D

daemon threads; 165

data

See also state;

contention avoidance

and scalability; 237

hiding

thread-safety use; 16

nonatomic

64-bit operations; 36

sharing; 33–54

See also page renderer examples;

access coordination; 277–290, 319

advantages of threads; 2

shared data models; 198–202

synchronization costs; 8

split data models; 201, 201–202

stale; 35–36

versioned data model; 201

data race; 341

race condition vs.; 20_{fn}

data structure(s)

See also collections; object(s);

queue(s); stack(s); trees;

handling

See atomic/atomicity; confine-

ment; encapsulation; itera-

tors/iteration; recursion;

protection

and interruption handling; 142

shared

as serialization source; 226

testing insertion and removal han-

dling; 248

database(s)

deadlock recovery capabilities; 206

JDBC Connection

thread confinement use; 43

thread pool size impact; 171

Date

effectively immutable use; 53

dead-code elimination

and performance testing; 269–270

deadline-based waits

as feature of Condition; 307

deadlock(s); 205, 205–217

See also concurrent/concurrency,
errors; liveness; safety;

analysis

thread dump use; 216–217

as liveness failure; 8

avoidance

and thread confinement; 43_{fn}

nonblocking algorithm advan-

tages; 319, 329

strategies for; 215–217

cooperating objects; 211

diagnosis

strategies for; 215–217

dynamic lock order; 207–210

in GUI framework; 190

lock splitting as risk factor for; 235

locking during iteration risk of; 83

recovery

database capabilities; 206

polled and timed lock acqui-

sition use; 279, 280

timed locks use; 215

reentrancy avoidance of; 27

resource; 213–215

thread starvation; 169, 168–169, 215

deadly embrace

See deadlock;

death, thread

abnormal, handling; 161–163

debugging

See also analysis; design; documenta-
tion; recovery; testing;

annotation use; 353

concurrency

costs vs. performance optimiza-
tion value; 224

custom thread factory as aid for; 175

JVM optimization pitfalls; 38_{fn}

thread dump use; 216_{fn}

thread dumps

intrinsic lock advantage over

ReentrantLock; 285–286

unbounded thread creation risks;

decomposition

See also composition; delegation;
encapsulation;
producer-consumer pattern; 89
tasks-related; 113–134

Decorator pattern

collection class use for wrapper factories; 60

decoupling

of activities
as producer-consumer pattern
advantage; 87
task decomposition as representation of; 113
of interrupt notification from handling in Thread interruption handling methods; 140
task submission from execution
and Executor framework; 117

delayed tasks

See also time/timing;
handling of; 123

DelayQueue

time management; 123

delegation

See also composition; design; safety;
advantages
class extension vs.; 314
for class maintenance safety; 234
thread safety; 234
failure causes; 67–68
management; 62

dependencies

See also atomic/atomicity; invariant(s); postconditions; preconditions; state;
code
as removal, as producer-consumer pattern advantage; 87
in multiple-variable invariants
thread safety issues; 24
state
blocking operations; 291–308
classes; 291
classes, building; 291–318
managing; 291–298
operations; 57
operations, condition queue handling; 296–308

task freedom from, importance of; 113

task

and execution policy; 167
thread starvation deadlock; 168
task freedom from
importance; 113

Deque; 92**deques**

See also collections; data structure(s);
queue(s);
work stealing and; 92

design

See also documentation; guidelines;
policies; representation;
strategies;

class

state ownership as element of;
57–58

concurrency design rules; 110

concurrency testing; 250–252

condition queue encapsulation; 306

condition queues

and condition predicate; 299

control flow

latch characteristics; 94

execution policy

influencing factors; 167

GUI single-threaded use

rationale for; 189–190

importance

in thread-safe programs; 16

of thread-safe classes

guidelines; 55–58

parallelism

application analysis for; 123–133

parallelization criteria; 181

performance

analysis, monitoring, and improvement; 221–245

performance tradeoffs

evaluation of; 223–225

principles

simplicity of final fields; 48

producer-consumer pattern

decoupling advantages; 117

Executor framework use; 117

program

and task decomposition; 113–134

result-bearing tasks

representation issues; 125

- strategies
 - for InterruptedException; 93
- thread confinement; 43
- thread pool size
 - relevant factors for; 170
- timed tasks; 131–133
- tradeoffs
 - collection copying vs. locking during iteration; 83
 - concurrent vs. synchronized collections; 85
 - copy-on-write collections; 87
 - synchronized block; 34
 - timeliness vs. consistency; 62, 66, 70
- design patterns**
 - antipattern example
 - double-checked locking; 348–349
 - examples
 - See Decorator pattern; MVC (model-view-controller) pattern; producer-consumer pattern; Singleton pattern;
- destruction**
 - See teardown;
- dining philosophers problem**; 205
 - See also deadlock;
- discard saturation policy**; 174
- discard-oldest saturation policy**; 174
- documentation**
 - See also debugging; design; good practices; guidelines; policy(s);
 - annotation use; 6, 353
 - concurrency design rules role; 110
 - critical importance for conditional notification use; 304
 - importance
 - for special execution policy requirements; 168
 - stack confinement usage; 45
 - of synchronization policies; 74–77
 - safe publication requirements; 54
- double-checked locking (DCL)**; 348–349
 - as concurrency bug pattern; 272
- downgrading**
 - read-write lock implementation strategy; 287

- driver program**
 - for TimedPutTakeTest example; 262
- dynamic**
 - See also responsiveness;
 - compilation
 - as performance testing pitfall; 267–268
 - lock order deadlocks; 207–210

E

- EDT (event dispatch thread)**
 - GUI frameworks use; 5
 - single-threaded GUI use; 189
 - thread confinement use; 42
- Effective Java Programming Language Guide*; 46–48, 73, 166, 257, 292, 305, 314, 347
- efficiency**
 - See also performance;
 - responsiveness vs.
 - polling frequency; 143
 - result cache, building; 101–109
- elision**
 - lock; 231_{fn}
 - JVM optimization; 286
- encapsulation**
 - See also access; atomic/atomicity; confinement; safety; state; visibility;
 - breaking
 - costs of; 16–17
 - code
 - as producer-consumer pattern advantage; 87
 - composition use; 74
 - concurrency design rules role; 110
 - implementation
 - class extension violation of; 71
 - instance confinement relationship with; 58–60
 - invariant management with; 44
 - locking behavior
 - reentrancy facilitation of; 27
 - non-standard cancellation; 148–150
 - of condition queues; 306
 - of lifecycle methods; 155
 - of synchronization
 - hidden iterator management through; 83
 - publication dangers for; 39
 - state

- breaking, costs of; 16–17
 - invariant protection use; 83
 - ownership relationship with; 58
 - synchronizer role; 94
 - thread-safe class use; 23
- synchronization policy
 - and client-side locking; 71
- thread ownership; 150
- thread-safety role; 55
- thread-safety use; 16
- end-of-lifecycle**
 - See also* thread(s);
 - management techniques; 135–166
- enforcement**
 - locking policies, lack of; 28
- entry protocols**
 - state-dependent operations; 306
- Error**
 - Callable handling; 97
- error(s)**
 - as cancellation reason; 136
 - concurrency
 - See* deadlock; livelock; race conditions;
- escape; 39**
 - analysis; 230
 - prevention
 - in instance confinement; 59
 - publication and; 39–42
 - risk factors
 - in instance confinement; 60
- Ethernet protocol**
 - exponential backoff use; 219
- evaluation**
 - See also* design; measurement; testing;
 - of performance tradeoffs; 223–225
- event(s); 191**
 - as cancellation reason; 136
 - dispatch thread
 - GUI frameworks use; 5
 - handling
 - control flow, simple; 194_{fg}
 - model-view objects; 195_{fg}
 - threads benefits for; 4
 - latch handling based on; 99
 - main event loop
 - vs. event dispatch thread; 5
 - notification
 - copy-on-write collection advantages; 87
 - sequential processing
 - in GUI applications; 191
 - timing
 - and liveness failures; 8
- example classes**
 - AtomicPseudoRandom; 327_{li}
 - AttributeStore; 233_{li}
 - BackgroundTask; 199_{li}
 - BarrierTimer; 261_{li}
 - BaseBoundedBuffer; 293_{li}
 - BetterAttributeStore; 234_{li}
 - BetterVector; 72_{li}
 - Big; 258_{li}
 - BoundedBuffer; 248, 249_{li}, 297, 298_{li}
 - BoundedBufferTest; 250_{li}
 - BoundedExecutor; 175
 - BoundedHashSet; 100_{li}
 - BrokenPrimeProducer; 139_{li}
 - CachedFactorizer; 31_{li}
 - CancellableTask; 151_{li}
 - CasCounter; 323_{li}
 - CasNumberRange; 326_{li}
 - CellularAutomata; 102_{li}
 - Computable; 103_{li}
 - ConcurrentPuzzleSolver; 186_{li}
 - ConcurrentStack; 331_{li}
 - ConditionBoundedBuffer; 308, 309_{li}
 - Consumer; 256_{li}
 - Counter; 56_{li}
 - CountingFactorizer; 23_{li}
 - CrawlerThread; 157_{li}
 - DelegatingVehicleTracker; 65_{li}, 201
 - DemonstrateDeadlock; 210_{li}
 - Dispatcher; 212_{li}, 214_{li}
 - DoubleCheckedLocking; 349_{li}
 - ExpensiveFunction; 103_{li}
 - Factorizer; 109_{li}
 - FileCrawler; 91_{li}
 - FutureRenderer; 128_{li}
 - GrumpyBoundedBuffer; 292, 294_{li}
 - GuiExecutor; 192, 194_{li}
 - HiddenIterator; 84_{li}
 - ImprovedList; 74_{li}
 - Indexer; 91_{li}
 - IndexerThread; 157_{li}
 - IndexingService; 156_{li}
 - LazyInitRace; 21_{li}
 - LeftRightDeadlock; 207_{li}
 - LifecycleWebServer; 122_{li}
 - LinkedQueue; 334_{li}

ListHelper; 73, 74_{li}
 LogService; 153, 154_{li}
 LogWriter; 152_{li}
 Memoizer; 103_{li}, 108_{li}
 Memoizer2; 104_{li}
 Memoizer3; 106_{li}
 MonitorVehicleTracker; 63_{li}
 MutableInteger; 36_{li}
 MutablePoint; 64_{li}
 MyAppThread; 177, 178_{li}
 MyThreadFactory; 177_{li}
 Node; 184_{li}
 NoVisibility; 34_{li}
 NumberRange; 67_{li}
 OneShotLatch; 313_{li}
 OneValueCache; 49_{li}, 51_{li}
 OutOfTime; 124_{li}, 161
 PersonSet; 59_{li}
 Point; 64_{li}
 PossibleReordering; 340_{li}
 Preloader; 97_{li}
 PrimeGenerator; 137_{li}
 PrimeProducer; 141_{li}
 PrivateLock; 61_{li}
 Producer; 256_{li}
 PutTakeTest; 255_{li}, 260
 Puzzle; 183_{li}
 PuzzleSolver; 188_{li}
 QueueingFuture; 129_{li}
 ReaderThread; 149_{li}
 ReadWriteMap; 288_{li}
 ReentrantLockPseudoRandom; 327_{li}
 Renderer; 130_{li}
 SafeListener; 42_{li}
 SafePoint; 69_{li}
 SafeStates; 350_{li}
 ScheduledExecutorService; 145_{li}
 SemaphoreOnLock; 310_{li}
 Sequence; 7_{li}
 SequentialPuzzleSolver; 185_{li}
 ServerStatus; 236_{li}
 SimulatedCAS; 322_{li}
 SingleThreadRenderer; 125_{li}
 SingleThreadWebServer; 114_{li}
 SleepyBoundedBuffer; 295, 296_{li}
 SocketUsingTask; 151_{li}
 SolverTask; 186_{li}
 StatelessFactorizer; 18_{li}
 StripedMap; 238_{li}
 SwingUtilities; 191, 192, 193_{li}
 Sync; 343_{li}

SynchronizedFactorizer; 26_{li}
 SynchronizedInteger; 36_{li}
 TaskExecutionWebServer; 118_{li}
 TaskRunnable; 94_{li}
 Taxi; 212_{li}, 214_{li}
 TestHarness; 96_{li}
 TestingThreadFactory; 258_{li}
 ThisEscape; 41_{li}
 ThreadDeadlock; 169_{li}
 ThreadGate; 305_{li}
 ThreadPerTaskExecutor; 118_{li}
 ThreadPerTaskWebServer; 115_{li}
 ThreeStooges; 47_{li}
 TimedPutTakeTest; 261
 TimingThreadPool; 180_{li}
 TrackingExecutorService; 159_{li}
 UEHLogger; 163_{li}
 UnsafeCachingFactorizer; 24_{li}
 UnsafeCountingFactorizer; 19_{li}
 UnsafeLazyInitialization; 345_{li}
 UnsafeStates; 40_{li}
 ValueLatch; 184, 187_{li}
 VisualComponent; 66_{li}
 VolatileCachedFactorizer; 50_{li}
 WebCrawler; 160_{li}
 Widget; 27_{li}
 WithinThreadExecutor; 119_{li}
 WorkerThread; 227_{li}

exceptions

See also error(s); interruption; lifecycle;
 and precondition failure; 292–295
 as form of completion; 95
 Callable handling; 97
 causes
 stale data; 35
 handling
 Runnable limitations; 125
 logging
 UEHLogger example; 163_{li}
 thread-safe class handling; 82
 Timer disadvantages; 123
 uncaught exception handler; 162–
 163
 unchecked
 catching, disadvantages; 161

Exchanger

See also producer-consumer pattern;
 as two-party barrier; 101
 safe publication use; 53

execute

submit vs., uncaught exception handling; 163

execution

policies

design, influencing factors; 167
 Executors factory methods; 171
 implicit couplings between tasks
 and; 167–170
 parallelism analysis for; 123–133

task; 113–134

policies; 118–119
 sequential; 114

ExecutionException

Callable handling; 98

Executor framework; 117_{li}, 117–133

and GUI event processing; 191, 192
 and long-running GUI tasks; 195
 as producer-consumer pattern; 88
 execution policy design; 167
 FutureTask use; 97
 GuiExecutor example; 194_{li}
 single-threaded
 deadlock example; 169_{li}

ExecutorCompletionService

in page rendering example; 129

Executors

factory methods
 thread pool creation with; 120

ExecutorService

and service shutdown; 153–155
 cancellation strategy using; 146
 checkMail example; 158
 lifecycle methods; 121_{li}, 121–122

exhaustion

See failure; leakage; resource exhaustion;

exit protocols

state-dependent operations; 306

explicit locks; 277–290

interruption during acquisition; 148

exponential backoff

and avoiding livelock; 219

extending

existing thread-safe classes
 and client-side locking; 73
 strategies and risks; 71
 ThreadPoolExecutor; 179

external locking; 73**F****factory(s)**

See also creation;

methods

constructor use with; 42
 newTaskFor; 148
 synchronized collections; 79, 171
 thread pool creation with; 120
 thread; 175, 175–177

fail-fast iterators; 82

See also iteration/iterators;

failure

See also exceptions; liveness, failure;
 recovery; safety;

causes

stale data; 35

graceful degradation

task design importance; 113

management techniques; 135–166

modes

testing for; 247–274

precondition

bounded buffer handling of; 292

propagation to callers; 292–295

thread

uncaught exception handlers;
 162–163

timeout

deadlock detection use; 215

fairness

See also responsiveness; synchronization;

as concurrency motivation; 1

fair lock; 283

nonfair lock; 283

nonfair semaphores vs. fair

performance measurement; 265

queuing

intrinsic condition queues; 297_{fi}

ReentrantLock options; 283–285

ReentrantReadWriteLock; 287

scheduling

thread priority manipulation
 risks; 218

'fast path' synchronization

CAS-based operations vs.; 324
 costs of; 230

feedback*See also* GUI;

user

in long-running GUI tasks; 196_{ji}**fields**

atomic updaters; 335–336

hot fields

avoiding; 237

updating, atomic variable advantages; 239–240

initialization safety

final field guarantees; 48

FIFO queues

BlockingQueue implementations; 89

files*See also* data; database(s);

as communication mechanism; 1

final

and immutable objects; 48

concurrency design rules role; 110

immutability not guaranteed by; 47

safe publication use; 52

volatile vs.; 158_{fn}**finalizers**

JVM orderly shutdown use; 164

warnings; 165–166

finally block*See also* interruptions; lock(ing);

importance with explicit locks; 278

FindBugs code auditing tool*See also* tools;

as static analysis tool example; 271

locking failures detected by; 28_{fn}unreleased lock detector; 278_{fn}**fire-and-forget event handling strategy**

drawbacks of; 195

flag(s)*See* mutex;

cancellation request

as cancellation mechanism; 136

interrupted status; 138

flexibility*See also* responsiveness; scalability;

and instance confinement; 60

decoupling task submission from execution, advantages for; 119

immutable object design for; 47

in CAS-based algorithms; 322

interruption policy; 142

resource management

as blocking queue advantage; 88

task design guidelines for; 113

task design role; 113

flow control

communication networks, thread

pool comparison; 173_{fn}**fragility***See also* debugging; guidelines; robustness; safety; scalability; testing;

issues and causes

as class extension; 71

as client-side locking; 73

interruption use for non-

standard purposes; 138

issue; 43

piggybacking; 342

state-dependent classes; 304

volatile variables; 38

solutions

composition; 73

encapsulation; 17

stack confinement vs. ad-hoc

thread confinement; 44

frameworks*See also* AQS framework; data structure(s); Executor framework; RMI framework; Servlets framework;

application

and ThreadLocal; 46

serialization hidden in; 227

thread use; 9

thread use impact on applications; 9

threads benefits for; 4

functionality

extending for existing thread-safe classes

strategies and risks; 71

tests

vs. performance tests; 260

Future; 126_{ji}

cancellation

of long-running GUI task; 197

strategy using; 145–147

characteristics of; 95

encapsulation of non-standard cancellation use; 148

results handling capabilities; 125

safe publication use; 53

task lifecycle representation by; 125

task representation
implementation strategies; 126

FutureTask; 95

AQS use; 316
as latch; 95–98
completion notification
of long-running GUI task; 198
efficient and scalable cache implementation with; 105
example use; 97_{li}, 108_{li}, 151_{li}, 199_{li}
task representation use; 126

G

garbage collection

as performance testing pitfall; 266

gate

See also barrier(s); conditional;
latch(es);
as latch role; 94
ThreadGate example; 304

global variables

ThreadLocal variables use with; 45

good practices

See design; documentation; encapsulation; guidelines; performance; strategies;

graceful

degradation
and execution policy; 121
and saturation policy; 175
limiting task count; 119
task design importance; 113
shutdown
vs. abrupt shutdown; 153

granularity

See also atomic/atomicity; scope;
atomic variable advantages; 239–240
lock
Amdahl's law insights; 229
reduction of; 235–237
nonblocking algorithm advantages;
319
serialization
throughput impact; 228
timer
measurement impact; 264

guarded

objects; 28, 54
state
locks use for; 27–29

@GuardedBy; 353–354

and documenting synchronization
policy; 7_{fn}, 75

GUI (Graphical User Interface)

See also event(s); single-thread(ed);
Swing;
applications; 189–202
thread safety concerns; 10–11
frameworks
as single-threaded task execution example; 114_{fn}
long-running task handling; 195–198
MVC pattern use
in vehicle tracking example; 61
response-time sensitivity
and execution policy; 168
single-threaded use
rationale for; 189–190
threads benefits for; 5

guidelines

See also design; documentation; policy(s); strategies;
allocation vs. synchronization; 242
atomicity
definitions; 22
concurrency design rules; 110
Condition methods
potential confusions; 307
condition predicate
documentation; 299
lock and condition queue relationship; 300
condition wait usage; 301
confinement; 60
deadlock avoidance
alien method risks; 211
lock ordering; 206
open call advantages; 213
thread starvation; 169
documentation
value for safety; 16
encapsulation; 59, 83
value for safety; 16
exception handling; 163
execution policy
design; 119
special case implications; 168
final field use; 48
finalizer precautions; 166
happens-before use; 346
immutability

- effectively immutable objects; 53
- objects; 46
- requirements for; 47
- value for safety; 16
- initialization safety; 349, 350
- interleaving diagrams; 6
- interruption handling
 - cancellation relationship; 138
 - importance of interruption policy knowledge; 142, 145
 - interrupt swallowing precautions; 143
- intrinsic locks vs. `ReentrantLock`; 285
- invariants
 - locking requirements for; 29
 - thread safety importance; 57
 - value for safety; 16
- lock
 - contention, reduction; 233
 - contention, scalability impact; 231
 - holding; 32
 - ordering, deadlock avoidance; 206
- measurement
 - importance; 224
- notification; 303
- objects
 - stateless, thread-safety of; 19
- operation ordering
 - synchronization role; 35
- optimization
 - lock contention impact; 231
 - premature, avoidance of; 223
- parallelism analysis; 123–133
- performance
 - optimization questions; 224
 - simplicity vs.; 32
- postconditions; 57
- private field use; 48
- publication; 52, 54
- safety
 - definition; 18
 - testing; 252
- scalability; 84
 - attributes; 222
 - locking impact on; 232
- sequential loops
 - parallelization criteria; 181
- serialization sources; 227

- sharing
 - safety strategies; 16
- sharing objects; 54
- simplicity
 - performance vs.; 32
- starvation avoidance
 - thread priority precautions; 218
- state
 - consistency preservation; 25
 - managing; 23
 - variables, independent; 68
- stateless objects
 - thread-safety of; 19
- synchronization
 - immutable objects as replacement for; 52
 - shared state requirements for; 28
- task cancellation
 - criteria for; 147
- testing
 - effective performance tests; 270
 - timing-sensitive data races; 254
- this reference
 - publication risks; 41
- threads
 - daemon thread precautions; 165
 - handling encapsulation; 150
 - lifecycle methods; 150
 - pools; 174
 - safety; 18, 55
- volatile variables; 38

H

hand-over-hand locking; 282

happens-before

- JMM definition; 340–342
- piggybacking; 342–344
- publication consequences; 244–249

hardware

- See also* CPU utilization;
- concurrency support; 321–324
- JVM interaction
 - reordering; 34
- platform memory models; 338
- timing and ordering alterations by
 - thread safety risks; 7

hashcodes/hashtables

- See also* collections;
- `ConcurrentHashMap`; 84–86
 - performance advantages of; 242
- `Hashtable`; 79

- safe publication use; 52
- inducing lock ordering with; 208
- lock striping use; 237
- heap inspection tools**
 - See also* tools;
 - measuring memory usage; 257
- Heisenbugs**; 247_{fn}
- helper classes**
 - and extending class functionality; 72–73
- heterogeneous tasks**
 - parallelization limitations; 127–129
- hijacked signal**
 - See* missed signals;
- Hoare, C. A. R.**
 - Java monitor pattern inspired by (bibliographic reference); 60_{fn}
- hoisting**
 - variables
 - as JVM optimization pitfall; 38_{fn}
- homogeneous tasks**
 - parallelism advantages; 129
- hooks**
 - See also* extending;
 - completion
 - in `FutureTask`; 198
 - shutdown; 164
 - JVM orderly shutdown; 164–165
 - single shutdown
 - orderly shutdown strategy; 164
 - `ThreadPoolExecutor` extension; 179
- hot fields**
 - avoidance
 - scalability advantages; 237
 - updating
 - atomic variable advantages; 239–240
- HotSpot JVM**
 - dynamic compilation use; 267
- 'how fast'; 222**
 - See also* GUI; latency; responsiveness;
 - vs. 'how much'; 222
- 'how much'; 222**
 - See also* capacity; scalability;
 - throughput;
 - importance for server applications; 223
 - vs. 'how fast'; 222

HttpSession

- thread-safety requirements; 58_{fn}

I

I/O

- See also* resource(s);
- asynchronous
 - non-interruptible blocking; 148
- message logging
 - reduction strategies; 243–244
- operations
 - thread pool size impact; 170
- sequential execution limitations; 124
- server applications
 - task execution implications; 114
- synchronous
 - non-interruptible blocking; 148
 - threads use to simulate; 4
 - utilization measurement tools; 240

idempotence

- and race condition mitigation; 161

idioms

- See also* algorithm(s); conventions;
- design patterns; document-
- ation; policy(s); protocols;
- strategies;
- double-checked locking (DCL)
 - as bad practice; 348–349
- lazy initialization holder class; 347–348
- private constructor capture; 69_{fn}
- safe initialization; 346–348
- safe publication; 52–53

IllegalStateException

- `Callable` handling; 98

@Immutable; 7, 353

immutable/immutability; 46–49

- See also* atomic/atomicity; safety;
- concurrency design rules role; 110
- effectively immutable objects; 53
- initialization safety guarantees; 51
- initialization safety limitation; 350
- objects; 46
 - publication with volatile; 48–49
 - requirements for; 47
- role in synchronization policy; 56
- thread-safety use; 16

implicit coupling

- See also* dependencies;
- between tasks and execution poli-
- cies; 167–170

improper publication; 51*See also* safety;**increment operation (++)**

as non-atomic operation; 19

independent/independence; 25*See also* dependencies; encapsulation; invariant(s); state;

multiple-variable invariant lack of thread safety issues; 24

parallelization use; 183

state variables; 66, 66–67

lock splitting use with; 235

task

concurrency advantages; 113

inducing lock ordering

for deadlock avoidance; 208–210

initialization*See also* construction/constructors; lazy; 21

as race condition cause; 21–22

safe idiom for; 348_{li}

unsafe publication risks; 345

safety

and immutable objects; 51

final field guarantees; 48

idioms for; 346–348

JMM support; 349–350

inner classes

publication risks; 41

instance confinement; 59, 58–60*See also* confinement; encapsulation;**instrumentation***See also* analysis; logging; monitoring; resource(s), management; statistics; testing;

of thread creation

thread pool testing use; 258

potential

as execution policy advantage;

121

service shutdown use; 158

support

Executor framework use; 117

thread pool size requirements determination use of; 170

ThreadPoolExecutor hooks for; 179

interfaces

user

threads benefits for; 5

interleaving

diagram interpretations; 6

generating

testing use; 259

logging output

and client-side locking; 150_{fn}operation; 81_{fg}

ordering impact; 339

thread

dangers of; 5–8

timing dependencies impact on race conditions; 20

thread execution

in thread safety definition; 18

interrupted (Thread)

usage precautions; 140

InterruptedException

flexible interruption policy advantages; 142

interruption API; 138

propagation of; 143_{li}

strategies for handling; 93

task cancellation

criteria for; 147

interruption(s); 93, 135, 138–150*See also* completion; errors; lifecycle; notification; termination;

triggering;

and condition waits; 307

blocking and; 92–94

blocking test use; 251

interruption response strategy

exception propagation; 142

status restoration; 142

lock acquisition use; 279–281

non-cancellation uses for; 143

non-interruptable blocking

handling; 147–150

reasons for; 148

policies; 141, 141–142

preemptive

deprecation reasons; 135_{fn}

request

strategies for handling; 140

responding to; 142–150

swallowing

as discouraged practice; 93

bad consequences of; 140

when permitted; 143

thread; 138

volatile variable use with; 39

intransitivity

encapsulation characterized by; 150

intrinsic condition queues; 297

disadvantages of; 306

intrinsic locks; 25, 25–26

See also encapsulation; lock(ing);
safety; synchronization;
thread(s);

acquisition, non-interruptable blocking reason; 148

advantages of; 285

explicit locks vs.; 277–278

intrinsic condition queue relationship to; 297

limitations of; 28

recursion use; 237_{fn}

ReentrantLock vs.; 282–286

visibility management with; 36

invariant(s)

See also atomic/atomicity; post-conditions; pre-conditions;
state;

and state variable publication; 68

BoundedBuffer example; 250

callback testing; 257

concurrency design rules role; 110

encapsulation

state, protection of; 83

value for; 44

immutable object use; 49

independent state variables requirements; 66–67

multivariable

and atomic variables; 325–326

atomicity requirements; 57, 67–68

locking requirements for; 29

preservation of, as thread safety requirement; 24

thread safety issues; 24

preservation of

immutable object use; 46

mechanisms and synchronization policy role; 55–56

publication dangers for; 39

specification of

thread-safety use; 16

thread safety role; 17

iostat application

See also measurement; tools;

I/O measurement; 240

iterators/iteration

See also concurrent/concurrency;
control flow; recursion;

as compound action

in collection operations; 79

atomicity requirements during; 80

fail-fast; 82

ConcurrentModificationException exception with; 82–83

hidden; 83–84

locking

concurrent collection elimination
of need for; 85

disadvantages of; 83

parallel iterative algorithms

barrier management of; 99

parallelization of; 181

unreliable

and client-side locking; 81

weakly consistent; 85

J

Java Language Specification, The; 53,
218_{fn}, 259, 358

Java Memory Model (JMM); 337–352

See also design; safety; synchronization;
visibility;

initialization safety guarantees for
immutable objects; 51

Java monitor pattern; 60, 60–61

composition use; 74

vehicle tracking example; 61–71

Java Programming Language, The; 346

java.nio package

synchronous I/O

non-interruptable blocking; 148

JDBC (Java Database Connectivity)

Connection

thread confinement use; 43

JMM (Java Memory Model)

See Java Memory Model (JMM);

join (Thread)

timed

problems with; 145

JSPs (JavaServer Pages)

thread safety requirements; 10

JVM (Java Virtual Machine)

See also optimization;

deadlock handling limitations; 206

escape analysis; 230–231

lock contention handling; 320_{fn}

- nonblocking algorithm use; 319
- optimization pitfalls; 38_{fn}
- optimizations; 286
- service shutdown issues; 152–153
- shutdown; 164–166
 - and daemon threads; 165
 - orderly shutdown; 164
- synchronization optimization by; 230
- thread timeout interaction
 - and core pool size; 172_{fn}
- thread use; 9
- uncaught exception handling; 162_{fn}

K

keep-alive time

- thread termination impact; 172

L

latch(es); 94, 94–95

- See also* barriers; blocking; semaphores; synchronizers;
- barriers vs.; 99
- binary; 304
 - AQS-based; 313–314
- FutureTask; 95–98
- puzzle-solving framework use; 184
- ThreadGate example; 304

layering

- three-tier application
 - as performance vs. scalability illustration; 223

lazy initialization; 21

- as race condition cause; 21–22
- safe idiom for; 348_{li}
- unsafe publication risks; 345

leakage

- See also* performance;
- resource
 - testing for; 257
- thread; 161
 - Timer problems with; 123
 - UncaughtExceptionHandler prevention of; 162–163

lexical scope

- as instance confinement context; 59

library

- thread-safe collections
 - safe publication guarantees; 52

Life cellular automata game

- barrier use for computation of; 101

lifecycle

- See also* cancellation; completion; construction/constructors; Executor; interruption; shutdown; termination; thread(s); time/timing;

- encapsulation; 155

Executor

- implementations; 121–122
- management strategies; 135–166
- support
 - Executor framework use; 117

task

- and Future; 125
- Executor phases; 125
- thread
 - performance impact; 116
 - thread-based service management; 150

lightweight processes

- See* threads;

linked lists

- LinkedBlockingDeque; 92
- LinkedBlockingQueue; 89
 - performance advantages; 263
 - thread pool use of; 173–174
- LinkedList; 85
- Michael-Scott nonblocking queue; 332–335
- nonblocking; 330

List

- CopyOnWriteArrayList as concurrent collection for; 84, 86

listeners

- See also* event(s);
- action; 195–197
- Swing
 - single-thread rule exceptions; 190
- Swing event handling; 194

lists

- See also* collections;
- CopyOnWriteArrayList
 - safe publication use; 52
 - versioned data model use; 201
- LinkedList; 85
- List
 - CopyOnWriteArrayList as concurrent replacement; 84, 86

Little's law

lock contention corollary; 232_{fn}

livelock; 219, 219

See also concurrent/concurrency,
errors; liveness;
as liveness failure; 8

liveness

See also performance; responsiveness
failure;

causes

See deadlock; livelock; missed
signals; starvation;

failure

avoidance; 205–220

improper lock acquisition risk of; 61
nonblocking algorithm advantages;
319–336

performance and

in servlets with state; 29–32

safety vs.

See safety;

term definition; 8

testing

criteria; 248

thread safety hazards for; 8

local variables

See also encapsulation; state; vari-
ables;

for thread confinement; 43

stack confinement use; 44

locality, loss of

as cost of thread use; 8

Lock; 277_{li}, 277–282

and Condition; 307

interruptible acquisition; 148

timed acquisition; 215

lock(ing); 85

See also confinement; encapsulation;
@GuardedBy; safety; synchro-
nization;

acquisition

AQS-based synchronizer opera-
tions; 311–313

improper, liveness risk; 61

interruptible; 279–281

intrinsic, non-interruptible

blocking reason; 148

nested, as deadlock risk; 208

polled; 279

protocols, instance confinement
use; 60

reentrant lock count; 26

timed; 279

and instance confinement; 59

atomic variables vs.; 326–329

avoidance

immutable objects use; 49

building

AQS use; 311

client-side; 72–73, 73

and compound actions; 79–82

condition queue encapsulation

impact on; 306

stream class management; 150_{fn}

vs. class extension; 73

coarsening; 231

as JVM optimization; 286

impact on splitting synchronized

blocks; 235_{fn}

concurrency design rules role; 110

ConcurrentHashMap strategy; 85

ConcurrentModificationException

avoidance with; 82

condition variable and condition

predicate relationship; 308

contention

measurement; 240–241

reduction, guidelines; 233

reduction, impact; 211

reduction, strategies; 232–242

scalability impact of; 232

coupling; 282

cyclic locking dependencies

as deadlock cause; 205

disadvantages of; 319–321

double-checked

as concurrency bug pattern; 272

elision; 231_{fn}

as JVM optimization; 286

encapsulation of

reentrancy facilitation; 27

exclusive

alternative to; 239–240

alternatives to; 321

inability to use, as Concurrent-
HashMap disadvantage; 86

timed lock use; 279

explicit; 277–290

interruption during lock acquisi-
tion use; 148

granularity

Amdahl's law insights; 229

- reduction of; 235–237
- hand-over-hand; 282
- in blocking actions; 292
- intrinsic; 25, 25–26
 - acquisition, non-interruptable
 - blocking reason; 148
 - advantages of; 285
 - explicit locks vs.; 277–278
 - intrinsic condition queue relationship to; 297
 - limitations of; 28
 - private locks vs.; 61
 - recursion use; 237_{fn}
 - ReentrantLock vs., performance considerations; 282–286
- iteration
 - concurrent collection elimination
 - of need for; 85
 - disadvantages of; 83
- monitor
 - See* intrinsic locks;
- non-block-structured; 281–282
- nonblocking algorithms vs.; 319
- open calls
 - for deadlock avoidance; 211–213
- ordering
 - deadlock risks; 206–213
 - dynamic, deadlocks resulting from; 207–210
 - inconsistent, as multithreaded GUI framework problem; 190
- private
 - intrinsic locks vs.; 61
- protocols
 - shared state requirements for; 28
- read-write; 286–289
 - implementation strategies; 287
- reentrant
 - semantics; 26–27
 - semantics, ReentrantLock capabilities; 278
- ReentrantLock fairness options; 283–285
- release
 - in hand-over-hand locking; 282
 - intrinsic locking disadvantages; 278
 - preference, in read-write lock implementation; 287
- role
 - synchronization policy; 56
- scope
 - See also* lock(ing), granularity; narrowing, as lock contention reduction strategy; 233–235
- splitting; 235
 - Amdahl's law insights; 229
 - as lock granularity reduction strategy; 235
 - ServerStatus examples; 236_{li}
 - state guarding with; 27–29
- striping; 237
 - Amdahl's law insights; 229
 - ConcurrentHashMap use; 85
- stripping; 237
- thread dump information about; 216
- thread-safety issues
 - in servlets with state; 23–29
- timed; 215–216
- unreleased
 - as concurrency bug pattern; 272
- visibility and; 36–37
- volatile variables vs.; 39
- wait
 - and condition predicate; 299
- lock-free algorithms; 329**
- logging**
 - See also* instrumentation; exceptions
 - UEHLogger example; 163_{li}
 - service
 - as example of stopping a thread-based service; 150–155
 - thread customization example; 177
 - ThreadPoolExecutor hooks for; 179
- logical state; 58**
- loops/looping**
 - and interruption; 143
- M**
- main event loop**
 - vs. event dispatch thread; 5
- Map**
 - ConcurrentHashMap as concurrent replacement; 84
 - performance advantages; 242
 - atomic operations; 86
- maximum pool size parameter; 172**
- measurement**
 - importance for effective optimization; 224
 - performance; 222

- profiling tools; 225
 - lock contention; 240
- responsiveness; 264–266
- strategies and tools
 - profiling tools; 225
- ThreadPoolExecutor hooks for; 179
- memoization; 103**
 - See also* cache/caching;
- memory**
 - See also* resource(s);
 - barriers; 230, 338
 - depletion
 - avoiding request overload; 173
 - testing for; 257
 - thread-per-task policy issue; 116
 - models
 - hardware architecture; 338
 - JMM; 337–352
 - reordering
 - operations; 339
 - shared memory multiprocessors; 338–339
 - synchronization
 - performance impact of; 230–231
 - thread pool size impact; 171
 - visibility; 33–39
 - ReentrantLock effect; 277
 - synchronized effect; 33
- Michael-Scott nonblocking queue;** 332–335
- missed signals;** 301, 301
 - See also* liveness;
 - as single notification risk; 302
- model(s)/modeling**
 - See also* Java Memory Model (JMM); MVC (model-view-controller) design pattern; representation; views;
 - event handling
 - model-view objects; 195_{fg}
 - memory
 - hardware architecture; 338
 - JMM; 337–352
 - model-view-controller pattern
 - deadlock risk; 190
 - vehicle tracking example; 61
 - programming
 - sequential; 2
 - shared data
 - See also* page renderer examples; in GUI applications; 198–202
 - simplicity
 - threads benefit for; 3
 - split data models; 201, 201–202
 - Swing event handling; 194
 - three-tier application
 - performance vs. scalability; 223
 - versioned data model; 201
 - modification**
 - concurrent
 - synchronized collection problems with; 82
 - frequent need for
 - copy-on-write collection not suited for; 87
 - monitor(s)**
 - See also* Java monitor pattern;
 - locks
 - See* intrinsic locks;
 - monitoring**
 - See also* instrumentation; performance; scalability; testing; tools;
 - CPU utilization; 240–241
 - performance; 221–245
 - ThreadPoolExecutor hooks for; 179
 - tools
 - for quality assurance; 273
 - monomorphic call transformation**
 - JVM use; 268_{ft}
 - mpstat application;** 240
 - See also* measurement; tools;
 - multiple-reader, single-writer locking**
 - and lock contention reduction; 239
 - read-write locks; 286–289
 - multiprocessor systems**
 - See also* concurrent/concurrency;
 - shared memory
 - memory models; 338–339
 - threads use of; 3
 - multithreaded**
 - See also* safety; single-thread(ed); thread(s);
 - GUI frameworks
 - issues with; 189–190
 - multivariable invariants**
 - and atomic variables; 325–326
 - atomicity requirements; 57, 67–68
 - dependencies, thread safety issues; 24
 - locking requirements for; 29

- preservation of, as thread safety requirement; 24
- mutable; 15**
 - objects
 - safe publication of; 54
 - state
 - managing access to, as thread safety goal; 15
- mutexes (mutual exclusion locks); 25**
 - binary semaphore use as; 99
 - intrinsic locks as; 25
 - ReentrantLock capabilities; 277
- MVC (model-view-controller) pattern**
 - deadlock risks; 190
 - vehicle tracking example use of; 61
- N**
- narrowing**
 - lock scope
 - as lock contention reduction strategy; 233–235
- native code**
 - finalizer use and limitations; 165
- navigation**
 - as compound action
 - in collection operations; 79
- newTaskFor; 126_{li}**
 - encapsulating non-standard cancellation; 148
- nonatomic 64-bit operations; 36**
- nonblocking algorithms; 319, 329, 329–336**
 - backoff importance for; 231_{fn}
 - synchronization; 319–336
 - SynchronousQueue; 174_{fn}
 - thread-safe counter use; 322–324
- nonfair semaphores**
 - advantages of; 265
- notification; 302–304**
 - See also* blocking; condition, queues; event(s); listeners; notify; notifyAll; sleeping; wait(s); waking up;
 - completion
 - of long-running GUI task; 198
 - conditional; 303
 - as optimization; 303
 - use; 304_{li}
 - errors
 - as concurrency bug pattern; 272
 - event notification systems

- copy-on-write collection advantages; 87
- notify**
 - as optimization; 303
 - efficiency of; 298_{fn}
 - missed signal risk; 302
 - notifyAll vs.; 302
 - subclassing safety issues
 - documentation importance; 304
 - usage guidelines; 303
- notifyAll**
 - notify vs.; 302
- @NotThreadSafe; 6, 353**
- NPTL threads package**
 - Linux use; 4_{fn}
- nulling out memory references**
 - testing use; 257
- O**
- object(s)**
 - See also* resource(s);
 - composing; 55–78
 - condition
 - explicit; 306–308
 - effectively immutable; 53
 - guarded; 54
 - immutable; 46
 - initialization safety; 51
 - publication using volatile; 48–49
 - mutable
 - safe publication of; 54
 - pools
 - appropriate uses; 241_{fn}
 - bounded, semaphore management of; 99
 - disadvantages of; 241
 - serial thread confinement use; 90
 - references
 - and stack confinement; 44
 - sharing; 33–54
 - state; 55
 - components of; 55
 - Swing
 - thread-confinement; 191–192
- objects**
 - guarded; 28
- open calls; 211, 211–213**
 - See also* encapsulation;
- operating systems**
 - concurrency use
 - historical role; 1

operations

- 64-bit, nonatomic nature of; 36
- state-dependent; 57

optimistic concurrency management

- See* atomic variables; CAS; nonblocking algorithms;

optimization

compiler

- as performance testing pitfall; 268–270

JVM

- pitfalls; 38_{fn}
- strategies; 286

lock contention

- impact; 231
- reduction strategies; 232–242

performance

- Amdahl's law; 225–229
- premature, avoidance of; 223
- questions about; 224
- scalability requirements vs.; 222

techniques

- See also* atomic variables; non-blocking synchronization;
- condition queues use; 297
- conditional notification; 303

order(ing)

- See also* reordering; synchronization;
- acquisition, in `ReentrantReadWriteLock`; 317_{fn}

checksums

- safety testing use; 253

FIFO

- impact of caller state dependence handling on; 294_{fn}

lock

- deadlock risks; 206–213
- dynamic deadlock risks; 207–210
- inconsistent, as multithreaded GUI framework problem; 190

operation

- synchronization role; 35

partial; 340_{fn}

- happens-before, JMM definition; 340–342
- happens-before, piggybacking; 342–344
- happens-before, publication consequences; 244–249

- performance-based alterations in thread safety risks; 7

total

- synchronization actions; 341

orderly shutdown; 164**OutOfMemoryError**

- unbounded thread creation risk; 116

overhead

- See also* CPU utilization; measurement; performance;

impact of

- See* performance; throughput;

reduction

- See* nonblocking algorithms; optimization; thread(s), pools;

sources

- See* blocking/blocks; contention; context switching; multithreaded environments; safety; suspension; synchronization; thread(s), lifecycle;

ownership

- shared; 58
- split; 58
- state
 - class design issues; 57–58
- thread; 150

P**page renderer examples**

- See also* model(s)/modeling, shared data;
- heterogeneous task partitioning; 127–129
- parallelism analysis; 124–133
- sequential execution; 124–127

parallelizing/parallelism

- See also* concurrent/concurrency; Decorator pattern;
- application analysis; 123–133
- heterogeneous tasks; 127–129
- iterative algorithms
 - barrier management of; 99
- puzzle-solving framework; 183–188
- recursive algorithms; 181–188
- serialization vs.
 - Amdahl's law; 225–229
- task-related decomposition; 113
- thread-per-task policy; 115

partial ordering; 340_{fn}

- happens-before
 - and publication; 244–249
 - JMM definition; 340

- piggybacking; 342–344
- partitioning**
 - as parallelizing strategy; 101
- passivation**
 - impact on HttpSession thread-safety requirements; 58_{fn}
- perfbar application**
 - See also* measurement; tools;
 - CPU performance measure; 261
 - performance measurement use; 225
- perfmon application; 240**
 - See also* measurement; tools;
 - I/O measurement; 240
 - performance measurement use; 230
- performance; 8, 221, 221–245**
 - See also* concurrent/concurrency;
 - liveness; scalability; throughput; utilization;
 - and heterogeneous tasks; 127
 - and immutable objects; 48_{fn}
 - and resource management; 119
 - atomic variables
 - locking vs.; 326–329
 - cache implementation issues; 103
 - composition functionality extension
 - mechanism; 74_{fn}
 - costs
 - thread-per-task policy; 116
 - fair vs. nonfair locking; 284
 - hazards
 - See also* overhead; priority(s), inversion;
 - JVM interaction with hardware
 - reordering; 34
 - liveness
 - in servlets with state; 29–32
 - locking
 - during iteration impact on; 83
 - measurement of; 222
 - See also* capacity; efficiency; latency; scalability; service time; throughput;
 - locks vs. atomic variables; 326–329
 - memory barrier impact on; 230
 - notifyAll impact on; 303
 - optimization
 - See also* CPU utilization; piggybacking;
 - Amdahl's law; 225–229
 - bad practices; 348–349
 - CAS-based operations; 323
 - reduction strategies; 232–242
- page renderer example with CompletionService
 - improvements; 130
- producer-consumer pattern advantages; 90
- read-write lock advantages; 286–289
- ReentrantLock vs. intrinsic locks; 282–286
- requirements
 - thread-safety impact; 16
- scalability vs.; 222–223
 - issues, three-tier application
 - model as illustration; 223
 - lock granularity reduction; 239
 - object pooling issues; 241
- sequential event processing; 191
- simplicity vs.
 - in refactoring synchronized blocks; 34
- synchronized block scope; 30
- SynchronousQueue; 174_{fn}
- techniques for improving
 - atomic variables; 319–336
 - nonblocking algorithms; 319–336
- testing; 247–274
 - criteria; 248
 - goals; 260
 - pitfalls, avoiding; 266–270
- thread pool
 - size impact; 170
 - tuning; 171–179
- thread safety hazards for; 8
- timing and ordering alterations for
 - thread safety risks; 7
- tradeoffs
 - evaluation of; 223–225
- permission**
 - codebase
 - and custom thread factory; 177
- permits; 98**
 - See also* semaphores;
- pessimistic concurrency management**
 - See* lock(ing), exclusive;
- piggybacking; 344**
 - on synchronization; 342–344
- point(s)**
 - barrier; 99
 - cancellation; 140

poison

- message; **219**
 - See also* livelock;
- pill; **155**, 155–156
 - See also* lifecycle; shutdown;
 - CrawlerThread; 157_{li}
 - IndexerThread; 157_{li}
 - IndexingService; 156_{li}
 - unbounded queue shutdown with; 155

policy(s)

- See also* design; documentation; guidelines; protocol(s); strategies;
- application
 - thread pool advantages; 120
- cancellation; **136**
 - for tasks, thread interruption policy relationship to; 141
 - interruption advantages as implementation strategy; 140
- execution
 - design, influencing factors; 167
 - Executors, for ThreadPoolExecutor configuration; 171
 - implicit couplings between tasks and; 167–170
 - parallelism analysis for; 123–133
 - task; 118–119
 - task, application performance importance; 113
- interruption; **141**, 141–142
- saturation; 174–175
- security
 - custom thread factory handling; 177
- sequential
 - task execution; 114
- sharing objects; 54
- synchronization; **55**
 - requirements, impact on class extension; 71
 - requirements, impact on class modification; 71
 - shared state requirements for; 28
- task scheduling
 - sequential; 114
 - thread pools; 117
 - thread pools advantages over thread-per-task; 121
 - thread-per-task; 115

thread confinement; 43

polling

- blocking state-dependent actions; 295–296
- for interruption; 143
- lock acquisition; 279

pool(s)

- See also* resource(s);
- object
 - appropriate uses; 241_{fn}
 - bounded, semaphore use; 99
 - disadvantages of; 241
 - serial thread confinement use; 90
- resource
 - semaphore use; 98–99
 - thread pool size impact; 171
- size
 - core; **171**, 172_{fn}
 - maximum; **172**
- thread; 119–121
 - adding statistics to; 179
 - application; 167–188
 - as producer-consumer design; 88
 - as thread resource management mechanism; 117
 - callback use in testing; 258
 - combined with work queues, in Executor framework; 119
 - configuration post-construction manipulation; 177–179
 - configuring task queue; 172–174
 - creating; 120
 - deadlock risks; 215
 - factory methods for; 171
 - sizing; 170–171
 - uncaught exception handling; 163

portal

- timed task example; 131–133

postconditions

- See also* invariant(s);
- preservation of
 - mechanisms and synchronization policy role; 55–56
- thread safety role; 17

precondition(s)

- See also* dependencies, state; invariant(s);
- condition predicate as; 299
- failure
 - bounded buffer handling of; 292

- propagation to callers; 292–295
- state-based
 - in state-dependent classes; 291
 - management; 57
- predictability**
 - See also* responsiveness;
 - measuring; 264–266
- preemptive interruption**
 - deprecation reasons; 135_{fn}
- presentation**
 - See* GUI;
- primitive**
 - local variables, safety of; 44
 - wrapper classes
 - atomic scalar classes vs.; 325
- priority(s)**
 - inversion; 320
 - avoidance, nonblocking algorithm advantages; 329
 - thread
 - manipulation, liveness hazards; 218
 - when to use; 219
- PriorityBlockingQueue**; 89
 - thread pool use of; 173–174
- PriorityQueue**; 85
- private**
 - constructor capture idiom; 69_{fn}
 - locks
 - Java monitor pattern vs.; 61
- probability**
 - deadlock avoidance use with timed and polled locks; 279
 - determinism vs.
 - in concurrent programs; 247
- process(es); 1**
 - communication mechanisms; 1
 - lightweight
 - See* threads;
 - threads vs.; 2
- producer-consumer pattern**
 - and Executor functionality
 - in CompletionService; 129
 - blocking queues and; 87–92
 - bounded buffer use; 292
 - control flow coordination
 - blocking queues use; 94
 - Executor framework use; 117
 - pathological waiting conditions; 300_{fn}
 - performance testing; 261
 - safety testing; 252
 - work stealing vs.; 92
- profiling**
 - See also* measurement;
 - JVM use; 320_{fn}
 - tools
 - lock contention detection; 240
 - performance measurement; 225
 - quality assurance; 273
- programming**
 - models
 - sequential; 2
- progress indication**
 - See also* GUI;
 - in long-running GUI task; 198
- propagation**
 - of interruption exception; 142
- protocol(s)**
 - See also* documentation; policy(s); strategies;
 - entry and exit
 - state-dependent operations; 306
 - lock acquisition
 - instance confinement use; 60
 - locking
 - shared state requirements for; 28
 - race condition handling; 21
 - thread confinement
 - atomicity preservation with open calls; 213
- pthreads (POSIX threads)**
 - default locking behavior; 26_{fn}
- publication; 39**
 - See also* confinement; documentation; encapsulation; sharing;
 - escape and; 39–42
 - improper; 51, 50–51
 - JMM support; 244–249
 - of immutable objects
 - volatile use; 48–49
 - safe; 346
 - idioms for; 52–53
 - in task creation; 126
 - of mutable objects; 54
 - serial thread confinement use; 90
 - safety guidelines; 49–54
 - state variables
 - safety, requirements for; 68–69
 - unsafe; 344–346

put-if-absent operation

See also compound actions;
 as compound action
 atomicity requirements; 71
 concurrent collection support for; 84

puzzle solving framework

as parallelization example; 183–188

Q**quality assurance**

See also testing;
 strategies; 270–274

quality of service

measuring; 264
 requirements
 and task execution policy; 119

Queue; 84–85**queue(s)**

See also data structures;
 blocking; 87–94
 cancellation, problems; 138
 cancellation, solutions; 140
 CompletionService as; 129
 producer-consumer pattern and;
 87–92
 bounded
 saturation policies; 174–175
 condition; 297
 blocking state-dependent operations use; 296–308
 intrinsic; 297
 intrinsic, disadvantages of; 306

FIFO; 89

implementations

 serialization differences; 227

priority-ordered; 89

synchronous

 design constraints; 89

 thread pool use of; 173

task

 thread pool use of; 172–174

unbounded

 poison pill shutdown; 156

using; 298

work

 in thread pools; 88, 119

R**race conditions; 7, 20–22**

See also concurrent/concurrency,
 errors; data, race; time/timing;
 avoidance
 immutable object use; 48
 in thread-based service shutdown; 153
 in GUI frameworks; 189
 in web crawler example
 idempotence as mitigating circumstance; 161

random(ness)

livelock resolution use; 219
 pseudorandom number generation
 scalability; 326–329
 test data generation use; 253

reachability

publication affected by; 40

read-modify-write operation

See also compound actions;
 as non-atomic operation; 20

read-write locks; 286–289**ReadWriteLock; 286_{li}**

exclusive locking vs.; 239

reaping

See termination;

reclosable thread gate; 304**recovery, deadlock**

See deadlock, recovery;

recursion

See also control flow; iterators/iteration;

intrinsic lock acquisition; 237_{fn}

parallelizing; 181–188

See also Decorator pattern;

reentrant/reentrancy; 26

and read-write locks; 287

locking semantics; 26–27

 ReentrantLock capabilities; 278

per-thread lock acquisition; 26–27

ReentrantLock; 277–282

ReentrantLock

AQS use; 314–315

intrinsic locks vs.

 performance; 282–286

Lock implementation; 277–282

random number generator using;

 327_{li}

Semaphore relationship with; 308

ReentrantReadWriteLock

AQS use; 316–317
 reentrant locking semantics; 287

references

stack confinement precautions; 44

reflection

atomic field updater use; 335

rejected execution handler

ExecutorService post-termination
 task handling; 121
 puzzle-solving framework; 187

RejectedExecutionException

abort saturation policy use; 174
 post-termination task handling; 122
 puzzle-solving framework use; 187

RejectedExecutionHandler

and saturation policy; 174

release

AQS synchronizer operation; 311
 lock
 in hand-over-hand locking; 282
 intrinsic locking disadvantages;
 278
 preferences in read-write lock
 implementation; 287
 unreleased lock bug pattern; 271
 permit
 semaphore management; 98

remote objects

thread safety concerns; 10

remove-if-equal operation

as atomic collection operation; 86

reordering; 34

See also deadlock; optimization; order(ing); ordering; synchronization; time/timing;

initialization safety limitation; 350
 memory

 barrier impact on; 230
 operations; 339

volatile variables warning; 38

replace-if-equal operation

as atomic collection operation; 86

representation

See also algorithm(s); design; documentation; state(s);

activities

 tasks use for; 113

algorithm design role; 104

result-bearing tasks; 125

task

 lifecycle, Future use for; 125

 Runnable use for; 125

 with Future; 126

 thread; 150

request

 interrupt

 strategies for handling; 140

requirements

See also constraints; design; documentation; performance;

concrete

 importance for effective performance optimization; 224

concurrency testing

 TCK example; 250

determination

 importance of; 223

independent state variables; 66–67

performance

 Amdahl's law insights; 229

 thread-safety impact; 16

synchronization

 synchronization policy component; 56–57

 synchronization policy documentation; 74–77

resource exhaustion, preventing

 bounded queue use; 173

 execution policy as tool for; 119

 testing strategies; 257

 thread pool sizing risks; 170

resource(s)

See also CPU; instrumentation; memory; object(s); pool(s); utilization;

accessing

 as long-running GUI task; 195

bound; 221

consumption

 thread safety hazards for; 8

deadlocks; 213–215

depletion

 thread-per-task policy issue; 116

increase

 scalability relationship to; 222

leakage

 testing for; 257

management

See also instrumentation; testing; dining philosophers problem;

- blocking queue advantages; 88
- execution policy as tool for; 119
- Executor framework use; 117
- finalizer use and limitations; 165
- graceful degradation, saturation
 - policy advantages; 175
- long-running task handling; 170
- saturation policies; 174–175
- single-threaded task execution
 - disadvantages; 114
- testing; 257
- thread pools; 117
- thread pools, advantages; 121
- thread pools, tuning; 171–179
- thread-per-task policy disadvantages; 116
- threads, keep-alive time impact
 - on; 172
- timed task handling; 131
- performance
 - analysis, monitoring, and improvement; 221–245
- pools
 - semaphore use; 98–99
 - thread pool size impact; 171
- utilization
 - Amdahl's law; 225
 - as concurrency motivation; 1
- response-time-sensitive tasks**
 - execution policy implications; 168
- responsiveness**
 - See also* deadlock; GUI; livelock; liveness; performance;
 - as performance testing criteria; 248
 - condition queues advantages; 297
 - efficiency vs.
 - polling frequency; 143
 - interruption policy
 - InterruptedException advantages; 142
 - long-running tasks
 - handling; 170
 - measuring; 264–266
 - page renderer example with CompletionService
 - improvements; 130
 - performance
 - analysis, monitoring, and improvement; 221–245
 - poor
 - causes and resolution of; 219
 - safety vs.
 - graceful vs. abrupt shutdown; 153
 - sequential execution limitations; 124
 - server applications
 - importance of; 113
 - single-threaded execution disadvantages; 114
 - sleeping impact on; 295
 - thread
 - pool tuning, ThreadPoolExecutor use; 171–179
 - request overload impact; 173
 - safety hazards for; 8
- restoring interruption status; 142**
- result(s)**
 - bearing latches
 - puzzle framework use; 184
 - cache
 - building; 101–109
 - Callable handling of; 125
 - Callable use instead of Runnable; 95
 - dependencies
 - task freedom from, importance of; 113
 - Future handling of; 125
 - handling
 - as serialization source; 226
 - irrelevance
 - as cancellation reason; 136, 147
 - non-value-returning tasks; 125
 - Runnable limitations; 125
- retry**
 - randomness, in livelock resolution; 219
- return values**
 - Runnable limitations; 125
- reuse**
 - existing thread-safe classes
 - strategies and risks; 71
- RMI (Remote Method Invocation)**
 - thread use; 9, 10
 - safety concerns and; 10
 - threads benefits for; 4
- robustness**
 - See also* fragility; safety;
 - blocking queue advantages; 88
 - InterruptedException advantages; 142
 - thread pool advantages; 120

rules

See also guidelines; policy(s); strategies;

happens-before; 341

Runnable

handling exceptions in; 143

task representation limitations; 125

running

ExecutorService state; 121

FutureTask state; 95

runtime

timing and ordering alterations by
thread safety risks; 7

RuntimeException

as thread death cause; 161

Callable handling; 98

catching

disadvantages of; 161

S**safety**

See also encapsulation; immutable
objects; synchronization;
thread(s), confinement;

cache implementation issues; 104

initialization

guarantees for immutable ob-
jects; 51

idioms for; 346–348

JMM support; 349–350

liveness vs.; 205–220

publication

idioms for; 52–53

in task creation; 126

of mutable objects; 54

responsiveness vs.

as graceful vs. abrupt shutdown;
153

split ownership concerns; 58

subclassing issues; 304

testing; 252–257

goals; 247

tradeoffs

in performance optimization
strategies; 223–224

untrusted code behavior

protection mechanisms; 161

saturation

policies; 174–175

scalability; 222, 221–245

algorithm

comparison testing; 263–264

Amdahl's law insights; 229

as performance testing criteria; 248

client-side locking impact on; 81

concurrent collections vs. synchro-
nized collections; 84

ConcurrentHashMap advantages; 85,
242

CPU utilization monitoring; 240–241

enhancement

reducing lock contention; 232–
242

heterogeneous task issues; 127

hot field impact on; 237

intrinsic locks vs. ReentrantLock

performance; 282–286

lock scope impact on; 233

locking during iteration risk of; 83

open call strategy impact on; 213

performance vs.; 222–223

lock granularity reduction; 239

object pooling issues; 241

three-tier application model as
illustration; 223

queue implementations

serialization differences; 227

result cache

building; 101–109

serialization impact on; 228

techniques for improving

atomic variables; 319–336

nonblocking algorithms; 319–336

testing; 261

thread safety hazards for; 8

under contention

as AQS advantage; 311

ScheduledThreadPoolExecutor

as Timer replacement; 123

scheduling

overhead

performance impact of; 222

priority manipulation risks; 218

tasks

sequential policy; 114

thread-per-task policy; 115

threads as basic unit of; 3

work stealing

dequeues and; 92

scope/scoped

- See also* granularity;
- containers
 - thread safety concerns; 10
- contention
 - atomic variable limitation of; 324
- escaping
 - publication as mechanism for; 39
- lock
 - narrowing, as lock contention reduction strategy; 233–235
- synchronized block; 30

search

- depth-first
 - breadth-first search vs.; 184
 - parallelization of; 181–182

security policies

- and custom thread factory; 177

Selector

- non-interruptable blocking; 148

semantics

- See also* documentation; representation;
- atomic arrays; 325
- binary semaphores; 99
- final fields; 48
- of interruption; 93
- of multithreaded environments
 - ThreadLocal variable considerations; 46
- reentrant locking; 26–27
 - ReentrantLock capabilities; 278
 - ReentrantReadWriteLock capabilities; 287
- undefined
 - of Thread.yield; 218
- volatile; 39
- weakly consistent iteration; 85
- within-thread-as-if-serial; 337

Semaphore; 98

- AQS use; 315–316
- example use; 100_{li}, 176_{li}, 249_{li}
- in BoundedBuffer example; 248
- saturation policy use; 175
- similarities to ReentrantLock; 308
- state-based precondition management with; 57

semaphores; 98, 98–99

- as coordination mechanism; 1
- binary
 - mutex use; 99

counting; 98

- permits, thread relationships; 248_{fn}
- SemaphoreOnLock example; 310_{li}
- fair vs. nonfair
 - performance comparison; 265
- nonfair
 - advantages of; 265

sendOnSharedLine example; 281_{li}**sequential/sequentiality**

- See also* concurrent/concurrency;
- asynchrony vs.; 2
- consistency; 338
- event processing
 - in GUI applications; 191
- execution
 - of tasks; 114
 - parallelization of; 181
- orderly shutdown strategy; 164
- page renderer example; 124–127
- programming model; 2
- task execution policy; 114
- tests, value in concurrency testing; 250
- threads simulation of; 4

serialized/serialization

- access
 - object serialization vs.; 27_{fn}
 - timed lock use; 279
 - WorkerThread; 227_{li}
- granularity
 - throughput impact; 228
- impact on HttpSession thread-safety requirements; 58_{fn}
- parallelization vs.
 - Amdahl's law; 225–229
- scalability impact; 228
- serial thread confinement; 90, 90–92
- sources
 - identification of, performance impact; 225

server

- See also* client;
- applications
 - context switch reduction; 243–244
 - design issues; 113

service(s)

- See also* applications; frameworks;
- logging

- as thread-based service example; 150–155
- shutdown
 - as cancellation reason; 136
- thread-based
 - stopping; 150–161
- servlets**
 - framework
 - thread safety requirements; 10
 - threads benefits for; 4
 - stateful, thread-safety issues
 - atomicity; 19–23
 - liveness and performance; 29–32
 - locking; 23–29
 - stateless
 - as thread-safety example; 18–19
- session-scoped objects**
 - thread safety concerns; 10
- set(s)**
 - See also* collection(s);
 - BoundedHashSet example; 100_{ii}
 - CopyOnWriteArraySet
 - as synchronized Set replacement; 86
 - safe publication use; 52
 - PersonSet example; 59_{ii}
 - SortedSet
 - ConcurrentSkipListSet as concurrent replacement; 85
 - TreeSet
 - ConcurrentSkipListSet as concurrent replacement; 85
- shared/sharing; 15**
 - See also* concurrent/concurrency; publication;
- data
 - See also* page renderer examples;
 - access coordination, explicit lock use; 277–290
 - models, GUI application handling; 198–202
 - synchronization costs; 8
 - threads advantages vs. processes; 2
- data structures
 - as serialization source; 226
- memory
 - as coordination mechanism; 1
- memory multiprocessors
 - memory models; 338–339
- mutable objects
 - guidelines; 54
- objects; 33–54
- split data models; 201–202
- state
 - managing access to, as thread safety goal; 15
- strategies
 - ExecutorCompletionService
 - use; 130
- thread
 - necessities and dangers in GUI applications; 189–190
 - volatile variables as mechanism for; 38
- shutdown**
 - See also* lifecycle;
 - abrupt
 - JVM, triggers for; 164
 - limitations; 158–161
 - as cancellation reason; 136
 - cancellation and; 135–166
 - ExecutorService state; 121
 - graceful vs. abrupt tradeoffs; 153
 - hooks; **164**
 - in orderly shutdown; 164–165
 - JVM; 164–166
 - and daemon threads; 165
 - of thread-based services; 150–161
 - orderly; **164**
 - strategies
 - lifecycle method encapsulation; 155
 - logging service example; 150–155
 - one-shot execution service example; 156–158
 - support
 - LifecycleWebServer example; 122_{ii}
- shutdown; 121**
 - logging service shutdown alternatives; 153
- shutdownNow; 121**
 - limitations; 158–161
 - logging service shutdown alternatives; 153
- side-effects**
 - as serialization source; 226
 - freedom from
 - importance for task independence; 113

- synchronized Map implementations
 - not available from Concurrent-HashMap; 86
- signal**
 - ConditionBoundedBuffer example; 308
- signal handlers**
 - as coordination mechanism; 1
- simplicity**
 - See also* design;
 - Java monitor pattern advantage; 61
 - of modeling
 - threads benefit for; 3
 - performance vs.
 - in refactoring synchronized blocks; 34
- simulations**
 - barrier use in; 101
- single notification**
 - See* notify; signal;
- single shutdown hook**
 - See also* hook(s);
 - orderly shutdown strategy; 164
- single-thread(ed)**
 - See also* thread(s); thread(s), confinement;
 - as Timer restriction; 123
 - as synchronization alternative; 42–46
 - deadlock avoidance advantages; 43_{fn}
 - subsystems
 - GUI implementation as; 189–190
 - task execution
 - disadvantages of; 114
 - executor use, concurrency prevention; 172, 177–178
- Singleton pattern**
 - ThreadLocal variables use with; 45
- size(ing)**
 - See also* configuration; instrumentation;
 - as performance testing goal; 260
 - bounded buffers
 - determination of; 261
 - heterogeneous tasks; 127
 - pool
 - core; 171, 172_{fn}
 - maximum; 172
 - task
 - appropriate; 113
 - thread pools; 170–171
- sleeping**
 - blocking state-dependent actions
 - blocking state-dependent actions; 295–296
- sockets**
 - as coordination mechanism; 1
 - synchronous I/O
 - non-interruptable blocking reason; 148
- solutions**
 - See also* interruption; results; search; termination;
- SortedMap**
 - ConcurrentSkipListMap as concurrent replacement; 85
- SortedSet**
 - ConcurrentSkipListSet as concurrent replacement; 85
- space**
 - state; 56
- specification**
 - See also* documentation;
 - correctness defined in terms of; 17
- spell checking**
 - as long-running GUI task; 195
- spin-waiting; 232, 295**
 - See also* blocking/blocks; busy-waiting;
 - as concurrency bug pattern; 273
- split(ing)**
 - data models; 201, 201–202
 - lock; 235
 - Amdahl's law insights; 229
 - as lock granularity reduction strategy; 235
 - ServerStatus examples; 236_{li}
 - ownership; 58
- stack(s)**
 - address space
 - thread creation constraint; 116_{fn}
 - confinement; 44, 44–45
 - See also* confinement; encapsulation;
 - nonblocking; 330
 - size
 - search strategy impact; 184
 - trace
 - thread dump use; 216
- stale data; 35–36**
 - improper publication risk; 51
 - race condition cause; 20_{fn}

starvation; 218, 218

See also deadlock; livelock; liveness; performance;

as liveness failure; 8

locking during iteration risk of; 83

thread starvation deadlock; **169**,
168–169

thread starvation deadlocks; **215**

state(s); 15

See also atomic/atomicity; encapsulation; lifecycle; representation; safety; visibility;

application

framework threads impact on; 9

code vs.

thread-safety focus; 17

dependent

classes; **291**

classes, building; 291–318

operations; **57**

operations, blocking strategies;
291–308

operations, condition queue handling; 296–308

operations, managing; 291

task freedom from, importance
of; 113

encapsulation

breaking, costs of; 16–17

invariant protection use; 83

synchronizer role; 94

thread-safe class use; 23

lifecycle

ExecutorService methods; 121

locks control of; 27–29

logical; **58**

management

AQS-based synchronizer operations; 311

managing access to

as thread safety goal; 15

modification

visibility role; 33

mutable

coordinating access to; 110

object; **55**

components of; 55

remote and thread safety; 10

ownership

class design issues; 57–58

servlets with

thread-safety issues, atomicity;
19–23

thread-safety issues, liveness
and performance concerns;
29–32

thread-safety issues, locking;
23–29

space; **56**

stateless servlet

as thread-safety example; 18–19

task

impact on Future.get; 95

intermediate, shutdown issues;
158–161

transformations

in puzzle-solving framework
example; 183–188

transition constraints; **56**

variables

condition predicate use; 299

independent; **66**, 66–67

independent, lock splitting; 235
safe publication requirements;
68–69

stateDependentMethod example; 301_{ii}

static

initializer

safe publication mechanism; 53,
347

static analysis tools; 271–273

statistics gathering

See also instrumentation;

adding to thread pools; 179

ThreadPoolExecutor hooks for; 179

status

flag

volatile variable use with; 38

interrupted; **138**

thread

shutdown issues; 158

strategies

See also design; documentation;

guidelines; policy(s); representation;

atomic variable use; 34

cancellation

Future use; 145–147

deadlock avoidance; 208, 215–217

delegation

vehicle tracking example; 64

design

- interruption policy; 93
 - documentation use
 - annotations value; 6
 - end-of-lifecycle management; 135–166
 - InterruptedException handling; 93
 - interruption handling; 140, 142–150
 - Future use; 146
 - lock splitting; 235
 - locking
 - ConcurrentHashMap advantages; 85
 - monitor
 - vehicle tracking example; 61
 - parallelization
 - partitioning; 101
 - performance improvement; 30
 - program design order
 - correctness then performance; 16
 - search
 - stack size impact on; 184
 - shutdown
 - lifecycle method encapsulation; 155
 - logging service example; 150–155
 - one-shot execution service example; 156–158
 - poison pill; 155–156
 - split ownership safety; 58
 - thread safety delegation; 234–235
 - thread-safe class extension; 71
- stream classes**
 - client-side locking with; 150_{fn}
 - thread safety; 150
- String**
 - immutability characteristics; 47_{fn}
- stripping**
 - See also* contention;
 - lock; 237, 237
 - Amdahl's law insights; 229
 - ConcurrentHashMap use; 85
- structuring**
 - thread-safe classes
 - object composition use; 55–78
- subclassing**
 - safety issues; 304
- submit, execute vs.**
 - uncaught exception handling; 163
- suspension, thread**
 - costs of; 232, 320
 - elimination by CAS-based concurrency mechanisms; 321
 - Thread.suspend, deprecation reasons; 135_{fn}
- swallowing interrupts**
 - as discouraged practice; 93
 - bad consequences of; 140
 - when permitted; 143
- Swing**
 - See also* GUI;
 - listeners
 - single-thread rule exceptions; 192
 - methods
 - single-thread rule exceptions; 191–192
 - thread
 - confinement; 42
 - confinement in; 191–192
 - use; 9
 - use, safety concerns and; 10–11
 - untrusted code protection mechanisms in; 162
- SwingWorker**
 - long-running GUI task support; 198
- synchronization/synchronized; 15**
 - See also* access; concurrent/concurrency; lock(ing); safety;;
 - allocation advantages vs.; 242
 - bad practices
 - double-checked locking; 348–349
 - blocks; 25
 - Java objects as; 25
 - cache implementation issues; 103
 - collections; 79–84
 - concurrent collections vs.; 84
 - problems with; 79–82
 - concurrent building blocks; 79–110
 - contended; 230
 - correctly synchronized program; 341
 - data sharing requirements for; 33–39
 - encapsulation
 - hidden iterator management through; 83
 - requirement for thread-safe classes; 18
 - 'fast path'
 - CAS-based operations vs.; 324
 - costs of; 230

- immutable objects as replacement; 52
- inconsistent
 - as concurrency bug pattern; 271
- memory
 - performance impact of; 230–231
- memory visibility use of; 33–39
- operation ordering role; 35
- piggybacking; 342–344
- policy; 55
 - documentation requirements; 74–77
- encapsulation, client-side locking violation of; 71
- race condition prevention with; 7
- requirements, impact on class extension; 71
- requirements, impact on class modification; 71
- shared state requirements for; 28
- ReentrantLock capabilities; 277
- requirements
 - synchronization policy component; 56–57
- thread safety need for; 5
- types
 - See* barriers; blocking, queues; FutureTask; latches; semaphores;
- uncontended; 230
- volatile variables vs.; 38
- wrapper
 - client-side locking support; 73
- synchronizedList (Collections)**
 - safe publication use; 52
- synchronizer(s); 94, 94–101**
 - See also* Semaphore; CyclicBarrier; FutureTask; Exchanger; CountdownLatch;
- behavior and interface; 308–311
- building
 - with AQS; 311
 - with condition queues; 291–318
- synchronous I/O**
 - non-interruptable blocking; 148
- SynchronousQueue; 89**
 - performance advantages; 174_{fn}
 - thread pool use of; 173, 174

T

- task(s); 113**
 - See also* activities; event(s); lifecycle;
- asynchronous
 - FutureTask handling; 95–98
- boundaries; 113
 - parallelism analysis; 123–133
 - using ThreadLocal in; 168
- cancellation; 135–150
 - policy; 136
 - thread interruption policy relationship to; 141
- completion
 - as cancellation reason; 136
 - service time variance relationship to; 264–266
- dependencies
 - execution policy implications; 167
 - thread starvation deadlock risks; 168
- execution; 113–134
 - in threads; 113–115
 - policies; 118–119
 - policies and, implicit couplings between; 167–170
 - policies, application performance importance; 113
 - sequential; 114
- explicit thread creation for; 115
- GUI
 - long-running tasks; 195–198
 - short-running tasks; 192–195
- heterogeneous tasks
 - parallelization limitations; 127–129
- homogeneous tasks
 - parallelism advantages; 129
- lifecycle
 - Executor phases; 125
 - ExecutorService methods; 121
 - representing with Future; 125
- long-running
 - responsiveness problems; 170
- parallelization of
 - homogeneous vs. heterogeneous; 129
- post-termination handling; 121
- queues
 - management, thread pool configuration issues; 172–174

- thread pool use of; 172–174
 - representation
 - Runnable use for; 125
 - with Future; 126
 - response-time sensitivity
 - and execution policy; 168
 - scheduling
 - thread-per-task policy; 115
 - serialization sources
 - identifying; 225
 - state
 - effect on Future.get; 95
 - intermediate, shutdown issues; 158–161
 - thread(s) vs.
 - interruption handling; 141
 - timed
 - handling of; 123
 - two-party
 - Exchanger management of; 101
- TCK (Technology Compatibility Kit)**
 - concurrency testing requirements; 250
- teardown**
 - thread; 171–172
- techniques**
 - See also* design; guidelines; strategies;
- temporary objects**
 - and ThreadLocal variables; 45
- terminated**
 - ExecutorService state; 121
- termination**
 - See also* cancellation; interruption; lifecycle;
 - puzzle-solving framework; 187
 - safety test
 - criteria for; 254, 257
 - thread
 - abnormal, handling; 161–163
 - keep-alive time impact on; 172
 - reasons for deprecation of; 135_{fn}
 - timed locks use; 279
- test example method**; 262_{li}
- testing**
 - See also* instrumentation; logging; measurement; monitoring; quality assurance; statistics;
 - concurrent programs; 247–274
 - deadlock risks; 210_{fn}
 - functionality
 - vs. performance tests; 260
 - liveness
 - criteria; 248
 - performance; 260–266
 - criteria; 248
 - goals; 260
 - pitfalls
 - avoiding; 266–270
 - dead code elimination; 269
 - dynamic compilation; 267–268
 - garbage collection; 266
 - progress quantification; 248
 - proving a negative; 248
 - timing and synchronization artifacts; 247
 - unrealistic code path sampling; 268
 - unrealistic contention; 268–269
 - program correctness; 248–260
 - safety; 252–257
 - criteria; 247
 - strategies; 270–274
- testPoolExample example**; 258_{li}
- testTakeBlocksWhenEmpty example**; 252_{li}
- this reference**
 - publication risks; 41
- Thread**
 - join
 - timed, problems with; 145
 - getState
 - use precautions; 251
 - interruption methods; 138, 139_{li}
 - usage precautions; 140
- thread safety**; 18, 15–32
 - and mutable data; 35
 - and shutdown hooks; 164
 - characteristics of; 17–19
 - data models, GUI application handling; 201
 - delegation; 62
 - delegation of; 234
 - in puzzle-solving framework; 183
 - issues, atomicity; 19–23
 - issues, liveness and performance; 29–32
 - mechanisms, locking; 23–29
 - risks; 5–8
- thread(s)**; 2
 - See also* concurrent/concurrency; safety; synchronization;

- abnormal termination of; 161–163
- as instance confinement context; 59
- benefits of; 3–5
- blocking; **92**
- confinement; **42**, 42–46
 - See also* confinement; encapsulation;
 - ad-hoc; 43
 - and execution policy; 167
 - in GUI frameworks; 190
 - in Swing; 191–192
 - role, synchronization policy specification; 56
 - stack; **44**, 44–45
 - ThreadLocal; 45–46
- cost
 - context locality loss; 8
 - context switching; 8
- costs; 229–232
- creation; 171–172
 - explicit creation for tasks; 115
 - unbounded, disadvantages; 116
- daemon; 165
- dumps; **216**
 - deadlock analysis use; 216–217
 - intrinsic lock advantage over ReentrantLock; 285
 - lock contention analysis use; 240
- factories; **175**, 175–177
- failure
 - uncaught exception handlers; 162–163
- forced termination
 - reasons for deprecation of; 135_{fn}
- interleaving
 - dangers of; 5–8
- interruption; **138**
 - shutdown issues; 158
 - status flag; **138**
- leakage; **161**
 - testing for; 257
 - Timer problems with; 123
 - UncaughtExceptionHandler prevention of; 162–163
- lifecycle
 - performance impact; 116
 - thread-based service management; 150
- overhead
 - in safety testing, strategies for mitigating; 254
- ownership; **150**
- pools; 119–121
 - adding statistics to; 179
 - and work queues; 119
 - application; 167–188
 - as producer-consumer design; 88
 - as thread resource management mechanism; 117
 - callback use in testing; 258
 - creating; 120
 - deadlock risks; 215
 - factory methods for; 171
 - post-construction configuration; 177–179
 - sizing; 170–171
 - task queue configuration; 172–174
- priorities
 - manipulation, liveness risks; 218
- priority
 - when to use; 219
- processes vs.; 2
- queued
 - SynchronousQueue management of; 89
- risks of; 5–8
- serial thread confinement; **90**, 90–92
- services that own
 - stopping; 150–161
- sharing
 - necessities and dangers in GUI applications; 189–190
- single
 - sequential task execution; 114
- sources of; 9–11
- starvation deadlock; **169**, 168–169
- suspension
 - costs of; 232, 320
 - Thread.suspend, deprecation reasons; 135_{fn}
- task
 - execution in; 113–115
 - scheduling, thread-per-task policy; 115
 - scheduling, thread-per-task policy disadvantages; 116
 - vs. interruption handling; 141
- teardown; 171–172
- termination
 - keep-alive time impact on; 172
- thread starvation deadlocks; **215**

thread-local

See also stack, confinement;
computation
role in accurate performance
testing; 268

Thread.stop

deprecation reasons; 135_{fn}

Thread.suspend

deprecation reasons; 135_{fn}

ThreadFactory; 176_{li}

customizing thread pool with; 175

ThreadInfo

and testing; 273

ThreadLocal; 45–46

and execution policy; 168
for thread confinement; 43
risks of; 46

ThreadPoolExecutor

and untrusted code; 162
configuration of; 171–179
constructor; 172_{li}
extension hooks; 179
newTaskFor; 126_{li}, 148

@ThreadSafe; 7, 353**throttling**

as overload management mechanism; 88, 173
saturation policy use; 174
Semaphore use in `BoundedExecutor`
example; 176_{li}

throughput

See also performance;
as performance testing criteria; 248
locking vs. atomic variables; 328
producer-consumer handoff
testing; 261
queue implementations
serialization differences; 227
server application
importance of; 113
server applications
single-threaded task execution
disadvantages; 114
thread safety hazards for; 8
threads benefit for; 3

Throwable

`FutureTask` handling; 98

time/timing

See also deadlock; lifecycle; order/ordering; race conditions;

-based task

handling; 123
management design issues; 131–133

barrier handling based on; 99

constraints

as cancellation reason; 136
in puzzle-solving framework;
187

interruption handling; 144–145

deadline-based waits

as feature of `Condition`; 307

deferred computations

design issues; 125

dynamic compilation

as performance testing pitfall;
267

granularity

measurement impact; 264

keep-alive

thread termination impact; 172

`LeftRightDeadlock` example; 207_{fg}

lock acquisition; 279

lock scope

narrowing, as lock contention
reduction strategy; 233–235

long-running GUI tasks; 195–198

long-running tasks

responsiveness problem handling; 170

measuring

in performance testing; 260–263
`ThreadPoolExecutor` hooks for;
179

performance-based alterations in
thread safety risks; 7

periodic tasks

handling of; 123

progress indication

for long-running GUI tasks; 198

relative vs. absolute

class choices based on; 123_{fn}

response

task sensitivity to, execution
policy implications; 168

short-running GUI tasks; 192–195

thread timeout

core pool size parameter impact
on; 172_{fn}

timed locks; 215–216

- weakly consistent iteration semantics; 86
 - TimeoutException**
 - in timed tasks; 131
 - task cancellation criteria; 147
 - Timer**
 - task-handling issues; 123
 - thread use; 9
 - timesharing systems**
 - as concurrency mechanism; 2
 - tools**
 - See also* instrumentation; measurement;
 - annotation use; 353
 - code auditing
 - locking failures detected by; 28_{fn}
 - heap inspection; 257
 - measurement
 - I/O utilization; 240
 - importance for effective performance optimization; 224
 - performance; 230
 - monitoring
 - quality assurance use; 273
 - profiling
 - lock contention detection; 240
 - performance measurement; 225
 - quality assurance use; 273
 - static analysis; 271–273
 - transactions**
 - See also* events;
 - concurrent atomicity similar to; 25
 - transformations**
 - state
 - in puzzle-solving framework example; 183–188
 - transition**
 - See also* state;
 - state transition constraints; 56
 - impact on safe state variable publication; 69
 - travel reservations portal example**
 - as timed task example; 131–133
 - tree(s)**
 - See also* collections;
 - models
 - GUI application handling; 200
 - traversal
 - parallelization of; 181–182
 - TreeMap**
 - ConcurrentSkipListMap as concurrent replacement; 85
 - TreeSet**
 - ConcurrentSkipListSet as concurrent replacement; 85
 - Treiber's nonblocking stack algorithm;** 331_{li}
 - trigger(ing)**
 - See also* interruption;
 - JVM abrupt shutdown; 164
 - thread dumps; 216
 - try-catch block**
 - See also* exceptions;
 - as protection against untrusted code behavior; 161
 - try-finally block**
 - See also* exceptions;
 - and uncaught exceptions; 163
 - as protection against untrusted code behavior; 161
 - tryLock**
 - barging use; 283_{fn}
 - deadlock avoidance; 280_{li}
 - trySendOnSharedLine example;** 281_{li}
 - tuning**
 - See also* optimization;
 - thread pools; 171–179
- ## U
- unbounded**
 - See also* bounded; constraints;
 - queue(s);
 - blocking waits
 - timed vs., in long-running task management; 170
 - queues
 - nonblocking characteristics; 87
 - poison pill shutdown use; 155
 - thread pool use of; 173
 - thread creation
 - disadvantages of; 116
 - uncaught exception handlers;** 162–163
 - See also* exceptions;
 - UncaughtExceptionHandler;** 163_{li}
 - custom thread class use; 175
 - thread leakage detection; 162–163
 - unchecked exceptions**
 - See also* exceptions;
 - catching
 - disadvantages of; 161

uncontended

synchronization; 230

unit tests

for BoundedBuffer example; 250
issues; 248

untrusted code behavior

See also safety;
ExecutorService code protection
strategies; 179
protection mechanisms; 161

updating

See also lifecycle;
atomic fields; 335–336
immutable objects; 47
views
in GUI tasks; 201

upgrading

read-write locks; 287

usage scenarios

performance testing use; 260

user

See also GUI;
cancellation request
as cancellation reason; 136
feedback
in long-running GUI tasks; 196_{li}
interfaces
threads benefits for; 5

utilization; 225

See also performance; resource(s);
CPU
Amdahl's law; 225, 226_{fg}
optimization, as multithreading
goal; 222
sequential execution limitations;
124
hardware
improvement strategies; 222

V**value(s)**

See result(s);

variables

See also encapsulation; state;
atomic
classes; 324–329
locking vs.; 326–329
nonblocking algorithms and;
319–336
volatile variables vs.; 39, 325–326
condition

explicit; 306–308
hoisting
as JVM optimization pitfall; 38_{fi}
local
stack confinement use; 44
multivariable invariant requirements
for atomicity; 57
state
condition predicate use; 299
independent; 66, 66–67
independent, lock splitting use
with; 235
object data stored in; 15
safe publication requirements;
68–69
ThreadLocal; 45–46
volatile; 38, 37–39
atomic variable class use; 319
atomic variable vs.; 39, 325–326
multivariable invariants prohib-
ited from; 68

variance

service time; 264

Vector

as safe publication use; 52
as synchronized collection; 79
check-then-act operations; 80_{li}, 79–
80
client-side locking management of
compound actions; 81_{li}

vehicle tracking example

delegation strategy; 64
monitor strategy; 61
state variable publication strategy;
69–71
thread-safe object composition de-
sign; 61–71

versioned data model; 201**views**

event handling
model-view objects; 195_{fg}
model-view-controller pattern
deadlock risks; 190
vehicle tracking example; 61
reflection-based
by atomic field updaters; 335
timeliness vs. consistency; 66, 70
updating
in long-running GUI task han-
dling; 201
with split data models; 201

visibility

See also encapsulation; safety; scope;
 condition queue
 control, explicit Condition and
 Lock use; 306
 guarantees
 JMM specification of; 338
 lock management of; 36–37
 memory; 33–39
 ReentrantLock capabilities; 277
 synchronization role; 33
 volatile reference use; 49

vmstat application

See also measurement; tools;
 CPU utilization measurement; 240
 performance measurement; 230
 thread utilization measurement; 241

Void

non-value-returning tasks use; 125

volatile

cancellation flag use; 136
 final vs.; 158_{fn}
 publishing immutable objects with;
 48–49
 safe publication use; 52
 variables; 38, 37–39
 atomic variable class use; 319
 atomic variable vs.; 39, 325–326
 atomicity disadvantages; 320
 multivariable invariants prohib-
 ited from; 68
 thread confinement use with; 43

W**wait(s)**

blocking
 timed vs. unbounded; 170
 busy-waiting; 295
 condition
 and condition predicate; 299
 canonical form; 301_{li}
 errors, as concurrency bug pat-
 tern; 272
 interruptible, as feature of Con-
 dition; 307
 uninterruptable, as feature of
 Condition; 307
 waking up from, condition
 queue handling; 300–301
 sets; 297

multiple, as feature of Condi-
 tion; 307
 spin-waiting; 232
 as concurrency bug pattern; 273
 waiting to run
 FutureTask state; 95

waking up

See also blocking/blocks; condition,
 queues; notify; sleep; wait;
 condition queue handling; 300–301

weakly consistent iterators; 85

See also iterators/iteration;

web crawler example; 159–161**within-thread usage**

See stack, confinement;

**within-thread-as-if-serial semantics;
337****work**

queues
 and thread pools, as producer-
 consumer design; 88
 in Executor framework use; 119
 thread pool interaction, size tun-
 ing requirements; 173
 sharing
 deque advantages for; 92
 stealing scheduling algorithm; 92
 deque and; 92
 tasks as representation of; 113

wrapper(s)

factories
 Decorator pattern; 60
 synchronized wrapper classes
 as synchronized collection
 classes; 79
 client-side locking support; 73